# FIVE PASSAGES TO THE FUTURE

KEMENTERIAN PENDIDIKAN DAN KEBUDAYAAN

GALERI NASIONAL INDONESIA

ARCOLABS

1999-2019

## CONTENT

## Foreword

### *FIVE PASSAGES TO THE FUTURE*

**Five Curators x Five Artists x Five Themes X Five Emerging Practices**

Artists today are often at the forefront in responding to numerous challenges in our society, and their art serves as a way of increasing a communal awareness of current problems and encouraging a collaborative effort to find solutions for the future. The notable challenges may include, but not limited to, energy shortage, ecological threat, climate change and science technology revolution called industry 4.0. Numerous efforts from various disciplines have been made to overcome these problems. One certain thing in solving these problems is that the long lasting idea of anthropocentrism (or human-centered) is no longer valid.

Five women curators with each distinguished expertise - art manager, artist, educator, writer, researcher - were invited to mount Five Passages to the Future exhibition in order to expose the most up-to-date technological aspects of new media art and its role in our society. The exhibition also aims to promote the practices of women curators in the new media art scene in Indonesia. Working closely with each artist, each curator proposed five cutting-edge approaches in the issues of eco-politics, sustainability, AI, digital narrative, and wearable technology, and each pair of curator-artist has collaboratively answered to the following fundamental questions:

- If humans and nature are equal components of a society, how can they co-exist?
  (Evelyn Huang X Digital Nativ)

- How can we pursue sustainable living on Earth?
  (Irene Agrivine X Eva Bubla)

- What is the role of humans in the age of artificial intelligence?
  (Jeong Ok Jeon X Cho Eun Woo)

- How does virtual reality as a new media change the human perception?
  (Nin Djani X Shezad Dawood)

- What kind of social change can we anticipate when fashion meets technology?
  (Ratna Djuwita X Synflux)



Five Passages to the Future also showcases diverse explorations of five emerging practices in the field of new media art who have participated in XPLORE: New Media Art Incubation program in 2018 and 2019, an education initiative by ARCOLABS and HONF. For this exhibition, the artists discussed ecology and data as its main themes, became more aware of the ecological shift towards a sustainable lifestyle and understood the essential aspect of data in making decision of what is possible in the future. By interpreting and elaborating data from our environments, the artists suggested how creative endeavor by new media art could contribute to our life.

Through collaborative works among various individuals, themes, and disciplines, this exhibition strives to show that human life becomes more meaningful when it builds relationships with a variety of entities - nature, artefacts, and technology in a symbiotic manner. Now, let us walk together through the five passages to embrace the future.

**Curatorial team**
**ARCOLABS & HONF Foundation**

# CHAPTER I
# ECOPOLITICS

## Curatorial Essay

### Art, Science and Nature: Towards Future Possibilities
### Evelyn Huang

Art and nature have always been in close relations. Since Plato's theory of mimesis, artistic practice has been continually involving nature - be it as object or phenomena – as a pivotal source of inspiration. In Post-Renaissance works, dialogues on art and nature was an intense discussion, as evident in the works of Rembrandt and Caravaggio, who applied the concept of chiaroscuro or depiction of light. Later, light was appreciated by a new generation of French painters who concentrated on presenting an impression of an object through the changing effect of light. This intellectual conversation with natural light prompted a new journey of modern Western art.

Art has also been associated with technology. Technology plays an important role in the development of human civilization. It is not a coincidence that the word "art" originates from the word "techne" (technique); technology has often been the driving force of new art movements such as the Italian Futurism and the German Bauhaus. Futurism celebrated the speed and dynamics of industrial machinery while Bauhaus promoted the fusion of art with technology as a cultural strategy to live in the modern era. Such technology-based products are the media.

Arguably, nature in art is slowly displaced by conversations of everyday life, embodied by technology itself. The everyday life (or technology) has been the main subject matter for many artworks that present ideas and cultures of the time, which we later call "the contemporary". Thereby, the function of nature changes from being the source of inspiration into mere objects, slowly losing its context. The approach to nature as a mere resource has changed the meaning of "exploration", as one of the most important processes in cultural practices, into "exploitation", occupied by transactional, capitalistic values. Accounts of destruction of nature such as human-made forest fire, oil spills, and global warming that pervade the conversations in our social space (or media) are reminder that nature actually remains as an important subject when we talk about the future of humanity.

Then, how does art talk about the future without losing the balance between nature and technology in this 21st century? Digital technology has made major changes in civilization, especially in art. The instant and replicative characteristics of digital technology enable tremendous acceleration in the process of production and cultural reproduction. In the context of fine arts, the influence of digital technology also makes it difficult to predict the direction of art. The birth of new media art is an example of this. As a recently developed format, new media art has established new artistic experiences for humans about space and time; about authentic, direct experiences and virtual artificial experiences. This innovative art practice is not just a reflection of digital media, but also seeks to become a projection in art and life. It builds new narratives about civilization for the future and creates new approaches to nature.

In the course of Indonesian art history, nature and technology too, have been an integral part. For example, *Mooi Indie* or Beautiful Indies, took off throughout the 19th century as a painting movement that romanticized the natural landscape of Indonesia. For the next two centuries, this style has continued to inspire generations of Indonesian artists to respond and examine the wider, historical and social context behind the paintings. Indeed, this practice of using art as more than a visual object/artwork but also a medium to convey political and social message is perhaps a unique Indonesian perspective of new media art. In reflecting above, the exhibition "Five Passages to the Future" places itself. Issues around sustainability, ecology as politics, new virtual reality, the challenge of artificial intelligence on humanity, and how the new narrative about technology itself become central topics to be discussed through new media art practices.

Human consciousness is connected with nature in various contexts. In general, nature is seen as a source of inspiration and also a resource. We are accustomed to see natural imitation (mimesis) as one of the artistic expressions in fine arts. On the other hand, natural exploration is seen as something that is extractive. What these two things have in common is our perspective in viewing nature as a powerless object. Efforts to conserve and restore nature / the environment often also take this perspective. One alternative that is developing in the realm of conservation is to throw away the perspective of human efforts to help nature as it is popular in the "save the earth" jargon, but to place natural strategies as the best solution for the balance of the ecosystem.

Through The Commons, Digital Nativ is trying to put nature as a subject by literally returning the microphone to the sound of nature. Departing from their field observations and research in the Katingan forest, Central Kalimantan, they were then involved in the process of peat forest conservation with the Katingan Mentaya Project. The experience of working together in a conservation ecosystem has added to their insight into the implementation of technology and art  in the conservation process. The Commons: Monoculture articulates a monoculture farming system focusing on the quantity and quality of crops that are usually achieved with the support of chemicals.



This method not only pollutes the crop, but also will kill the soil and damage the water in a short time, thus disturbing the balance of the ecosystem. The Commons: Biodiversity sets the spotlight for the unique biodiversity of tropical forests found in Indonesia. Both of these installations and their interactions with the audience present a scenario that is already in front of our eyes, a future that is not too far away to imagine but is sometimes overlooked in urban life. The artist put nature and social issues as the center of their artwork and as a result, it resonates with many audiences.

The conversation between art, science, technology and nature can be a new indicator for the direction and pace of art in the future. Not only do artists offer an expansion of sensory experiences conventionally provided by new media art, but they also highlight social and communal involvement. But no less important is how this whole discourse can manifest in actual discussions among the audience.

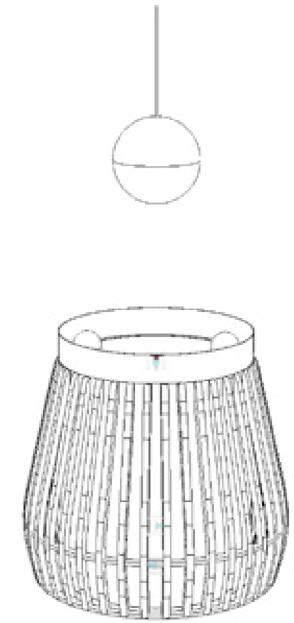
# Digital Nativ


## ARTIST STATEMENT

Consider this probable future, industrial agriculture continues to ignore the health of ecosystems in order to keep up with the demands of a growing population. We can assume how this trajectory plays out because we are already feeling its repercussions in the haze we breathe in and the climate we can no longer predict.

Now imagine a possible future in which equal value is given to ecosystems, their biodiversity valued as a natural resource belonging to the Commons. Inspired by the work of Katingan Mentaya Project in rehabilitating and conserving peatlands in South Kalimantan, 'The Commons' attempts to highlight the issues of land management in Indonesia, and to recognise the need for regenerative practices which increase biodiversity to be given equal value, for the good of our common future.





### MONOCULTURE

Feeding the future will be a momentous task. In the age of the anthropocene, ecological literacy will play a more important role than ever before and better knowledge of sustainable food production practices will be central to this.

Technological innovations aimed at understanding electrical signalling and the vital signs of plants may lead to optimization of environmental conditions and provide early warning of stress, disease or pest attack in plants. Alongside scalable hydroponics and new trends towards agroecology; ecological grazing, agroforestry, food forestry, biodynamics, perma-culture, cropping with biological inputs, there may be hope.

### BIODIVERSITY

The natural world is alive and filled with energy which we can sense in the form of electrical signals. Plants use these signals to communicate, grow, and respond to their environment. By capturing this biodata and converting the electrical signals into audible sound, we reveal a world usually hidden to our senses, giving nature a voice.

With this living exhibition, we invite you to interact with the plants and sensitise yourself to their responses. Listen to their 'song' and use the sensors to affect their musical fluctuations.

The Common: Biodiversity
Aglaonema, Asplenium, Calathea lancifolia, Ipomoea Batatas, Phalaenopsis Ambalis, Philodendron, Sanseviera Trifasciata, Syngonium Podophyllum, wood, acrylic, electronics, Nada Bumi system
70cm x 70cm x 85cm
2019

The Common: Monoculture
Oryza Sativa, acrylic, growlight, electronics, Nada Bumi system
30cm x 30cm x 200cm
2019



## *Critical Essay*

### The Commons: Ecology and Politics of Perceiving Nature
### Yuka Dian Narendra

1. Representation of Beautiful Realities: Seeing Nature through Art

Is it true that nature is considered to be a source of inspiration? The prevalent images of nature in art practice can strengthen the assumption that nature is a source of inspiration for artists, who incorporate nature as both a narrative, or subject matter in the work, and as materials. This assumption is not new actually; in the post-Renaissance Western cultural traditions it was the explosion of natural science that sparked conversations about the relationship between nature and humans. The same explosion also gave birth to the practice of painting natural objects, which became relevant when humans finally understood scientific aspects of nature. When these scientific aspects of nature became comprehensible to the human minds, painting about nature turned into an important practice to perceive nature. From this point forward, Western cultures had never stopped documenting natural and human phenomena. Eventually, this half-millennium-long practice arrived in Indonesia.

Those who attended elementary school throughout the ‘70s and ‘80s in Indonesia might have sat through art classes that asked them to draw the following scenery: a one-point perspective image of two mountains with a sun rising in between. There is a path leading to the midpoint of the two mountains just below the sun, flanked by either trees or power poles. On the sides of this path are green, luscious rice fields that look more like neatly assembled ceramic tiles. In the sky, twin lines in the shapes of “Golden Arches” decorate the image. Later, in the ‘90s, they would identify the arches as the McDonald’s logo, but in the picture these arches represent sparrows flying over the rice fields.

In the Indonesian art scene, such natural landscape pictures ignite interesting discussions. Some argue that the twin mountains in the picture are of Mount Sindoro and Sumbing in Central Java. Some insist that the practice has been culturally engineered by the Japanese colonial government, since the twin mountains actually symbolize Mount Fuji in Japan. Another stance on the debate is that the pictures are proof of a collective imagination among Indonesians about Java, because such natural landscapes only exist in Java. Regardless of which argument is correct, the conversations around this kind of landscape drawings lead to a critique of the dominant Javanese culture in the Indonesian national memory during the New Order.

In particular, the pictures indicate a Javanese hegemony that places its landscape as the paragon of Indonesian nature. The rich and diverse natural landscapes in Indonesia are reduced into a single trope that is the imaginary realm of Java.

The points stated above open a long discussion on the relationships of nature, reality, and aesthetics (or beauty). However, before we proceed, it is important to look at nature in the context of aesthetics. In philosophy, the intertwining nature and aesthetics is a discourse with two thousand years of history, beginning when Plato proposed the concept of mimesis or imitation. According to Plato, art is an imitation of nature, whereas nature represents reality. Albeit naively, we can ask whether what is represented by reality truly stands for humans. We can then interpret this view differently. When we assume that artists creatively produce "imitations" of nature, then what they really portray are representations of reality. Before we agree whether what the artists portray is beautiful, we need to consider the important question proposed by art historian E.H. Gombrich: do painters paint the world that they see, or, do painters see the world that they paint.

Beauty is a manifestation of the ideal world, a world of virtues envisioned by humans. Beauty represents the level of perfection. Such perfection refers to the human ideas or thoughts that convey virtues. In that case, the problem is how can we connect the two discourses? How do we link nature as a representation of reality and beauty as a manifestation of the human ideas of perfection? If Plato assumes that beauty exists only in the human minds, then nature loses its claim to beauty. As a representation, the role of nature in science is vital. When humans ask fundamental questions about their place in the world, nature offers an answer. The reality of nature in the physical sense is later referred to in science as "Earth", from which derives the natural sciences: physics, geology, astronomy, biology and chemistry. Meanwhile, the "World" comes to represent the relationships of humans, nature, and everything in between. The reality of nature in this social-humanities dimension provides an insight into how humans shape themselves and their culture. The interactions between humans and nature are ever-present dialogues in every culture, both in traditional and modern cultural practices.

In essence, art has been forever intertwined with nature, science and technology. Long since Plato proposed mimesis, art practice has always placed nature - both as objects and phenomena - as a source of inspiration. More than merely portraying beautiful natural sceneries, conversations around nature in art have actually been intense and filled with both contentions as well as admirations. This is evident in the attempts by post-Renaissance



European painters such as Rembrandt and Caravaggio who applied chiaroscuro or contrasting lights. Hundreds of years later, in the mid 19th century, once again light became a source of inspiration for European artists. At the time, many emerging painters truly recognized the impact of illumination levels on the quality of visible forms and colors of an object. This intellectual discussion gave birth to a new aesthetic movement we now recognize as Impressionism. From this point forward, the journey of modern Western art began.

In art practice, nature is considered as a scientific drawing object. One of its technical approaches is the perspective projection. Adopting the academic art tradition, natural landscapes are reconstructed using the perspective technique, which has geometrical and architectural bases. Mountains, rice fields, and oceans are approached in the same manner as urban landscapes. Meanwhile, many objects in nature, like weeds/plants, animals and humans are examined through biology. At the core of anatomical drawing is perspective projection as biology is used as methods of interpretation. What this approach can teach drawing apprentices like us is ways to understand the construction of nature and how nature is constructed in our minds. From there, we can begin to appreciate the basic visual aesthetics. It is from such beauty that we can approach nature and perceive it as the representation of a "beautiful" reality. Although, we are engrossed in studying the objective accuracy of the construction we draw, which happens to be natural objects.

2. Explorations and Extractions: Nature as Resource?

With this approach, the discourse of beautiful nature is actually set on the perfect construction of nature itself. Why? When we draw, don't we prioritize objective accuracy and overall precision? In art practice, nature is seen as a drawing object reproduced by artists through academic methods. It is through the development of science and technology that we agree on the reality of nature as the perfect reality that represents transcendental divine beauty. Then, practicing art becomes a practice of artistic expression that brings forth transcendental dimension. We can agree that in a beautiful landscape painting there are three components tied together.

First, the beautiful natural landscape that becomes the source of inspiration for the artists. Second, the artists' emotions towards the natural beauty they see. Third, the representations of nature in the form of paintings. Borrowing the question proposed by Gombrich, we can ask similar concerns. What is considered beautiful in art practice: nature, artistic expression, or the representation? If art practice is considered to be a practice of imitating nature and creating art is a form of artistic expressions by artists, then the practice of imitating nature is a divine artistic expression. This is because what is divine is natural, and so, what is natural must be divine. Yet, more than presenting nature to the public, artistic expressions can also present extra-aesthetic dimensions in nature. These are the dimensions that refer to other aspects of nature aside from its beauty, including biodiversity, geological information, as well as social issues around nature.

In parallel to the development of science and technology in the West, anthropocentrism emerged as a viewpoint that places humans as the center of civilization. When this became a new subject matter in the art practice (particularly in fine arts), the beautiful nature that was once glorified faded gradually and disappeared from the conversations. The fading and disappearing of nature from the artists' eyes is a logical consequence from the development of modern science and technology. Many science disciplines are only concerned about humans and their interests. This tendency has an impact in the marginalization of things that were previously considered sacred. Science once placed nature as the source of all essential questions regarding human existence, but now nature is shifted into mere research questions, reduced into formulas, and quantified as achievement probabilities or target results. The development of each branch of science has always been complex - and aligned with modernization and industrialization - thus demanding humans to become specialists in particular subjects. Such specialization of science isolates and alienates the subject from one another. As a result, each branch of science seems to have lost the ability to see the bigger picture of the fundamental question: what is their purpose in the context of civilization?

The fading and disappearing of nature from the artists' eye represents how we grow increasingly distant from the aforementioned senses of "the divine" and "the natural". Modernity has successfully built a new narrative about "what seems natural" in the daily lives of humans. This new narrative is none other than how science and technology constructs a world that is completely different from how it was several centuries prior, the old world that was created by humans themselves for millennia. This narrative seems to offer humans a new historical purpose: it is the human's duty to manage and optimize the world for their own prosperity. Here, science and technology become the instruments for humans to fulfill their duty. The emergence of cities, plantations, mines, factories, shopping centers, tourism sites, agriculture, farms, and many others are perceived as natural representations of developing civilizations and efforts to fulfill the needs of humans. Modifications of wildlife into economic objects are considered appropriate since these were only ways to fulfill the needs of humanity. This appropriation is then seen as natural. Over time, these fittingly



"natural" economic objects become a new natural reality for humans. Ownership of lands are now also applied to nature. The natural wilderness turns into empty lands to be explored entirely by humans.

The problem does not end there. Such situations affect the ways humans see nature. Aside from empty lands, nature is also viewed as a resource that can benefit the proprietors. This perspective quantifies the narrative about nature that once carried divine nuance (as the representation of beautiful and perfect reality) into economical abstraction. Nature can be valued and its depreciation can be calculated. In this context, depreciation only predicts how capital accumulation will be affected when nature loses its carrying capacity. More than that, nature can be valued as a source of raw materials that serve to bring profits. The economic valuations that fail to see nature as a representation of beautiful realities thus place nature and everything in it as mere objects that are free to conquer, manage, and exploit. Therefore, nature is no longer a source of inspiration, but a resource. Beauty no longer comes from nature because human nature has been filled with materiality. In science, nature is objectified by its own studies.

Beautiful, natural wilderness are replaced with urban landscapes. Religious and cultural rites turn into daily lives of urban dwellers. Natural objects like mountains, trees, and biodiversity change into hustling and bustling cities, skyscrapers, machines and other mechanical objects. Nature is no longer a source of inspiration, let alone source of beauty. The relevant meanings of nature have slipped since there are no more cultural practices attached to it. With science and technology, humans have successfully developed a new culture, in which its rites and practices are detached from nature and instead are intended solely for humans themselves (both as individuals and collectives). Because of its objectification, the reality of nature is treated merely as a background. As if it is a photography studio, nature serves a backdrop for the object of interest in front of the camera. If all the cultural phenomenon in this narrative is to be accumulated, then we can safely conclude that the highest achievement of modern science and technology is giving birth to a new civilization that is disconnected from nature. Nature does not only lose its meaning, but also its sacredness. Therefore the beauty of nature remains only as visual. This means that nature can only find its relevance when it is being appreciated like works of art, and not because of its important role in enabling life for all beings on earth. When nature becomes a mere visual, then its important role in this new civilization is nothing more than beautifying the daily lives of humans.

In the course of civilization, technology plays an important role. This role is more than an etymological imprint, which explains why the word "art" in the context of Western civilization comes from the word "techne" or technique. Referring to the modern art history in Europe, technology often becomes an inspiration for art movements. The Italian Futurism of 1909 states their artistic expression is a celebration of speed and the dynamics of industrial machines. Meanwhile in Germany, the Bauhaus (1919 - 1933) asserts that the integration of art and technology is a cultural strategy to live in the modern era.[i] The statement by Bauhaus has initiated modern design and integrations of many products of technology as part of both art practices and art objects. One of these products is the media. The media technology forms a path for humans to do what they have done for centuries: extracting nature as mere visuals. In this context, such visualization facilitated by the media enables nature to come and go as the users please, so that they get used to seeing nature and at the same time, watching it disappear in an instant.

The distant relationship between humans as cultured beings and their natural environment raises many problems. Especially when nature no longer has any value other than economical. In this perspective, explorations of nature that were initially intended to answer the needs of humans turn into extractive practices that lead to competitions for massive extractions. The creeds of industrialization are applied into methods of controlling nature. Although efficiency is the broad theme in this competition, such efficiency is only concerned about how the extractions are done, and not about nature. Humans, who in the beginning acted as proprietors/rulers/managers of natural resources become resources themselves - that we refer to as "human resources". Positioning humans as resources is closely tied to the capitalistic power structure. This means, those who can have access to large-scale natural resources generally can also have power over the human resources. In this context, the relation between owners of natural resources and human resources is a hegemonic one. Explorations and extractions of nature done by capitalism put nature as a powerless object of interest.

This particular perspective often results in exploitative explorations that affects the relations between humans and nature. Nature is objectified so much that it can be utilized (read: treated) as humans please. This objectification places humans in an "objective" distance from nature, where it is seen both as objects to explore economically and as objects of observation. As economic objects, the issue of utilizing natural resources frequently meets with the issue of conserving nature. Utilizing natural resources as economical means puts nature as a source of raw materials for production and therefore is closely linked to the livelihoods of the people. Before we realize it, "explorations" as an important cultural process turns into capitalistic exploitations. The narrative about destruction of nature that now pervades social spheres becomes a reminder that nature is



an important deciding factor in the conversations on the future of humanity. How this conversation unfolds leaves a big question for us: what is really at stake here?

3. Vox Natura Vox Dei: Nature as a Subject

From the long commentary that we have discussed in this essay, we can say the artistic practice of visualizing nature is inseparable from the way the discourse around nature is formed through science and state power. In this regard, visual art and music are major instruments that facilitate the state to produce a way of perceiving Indonesian nature. As stated by Rizki A. Zaelani (1999), "painting (natural) landscapes not only duplicates nature but also idealize it; it involves the artists' desire and agenda, which arguably are 'new' subjects produced by Indonesian nationalism."[ii] From this perspective, we can begin to understand how the beauty of Indonesian nature as painted in modern paintings provide a visual form - a face, in a way - for Indonesian nationalism. During the national revival movements, such visualization served to represent the locus of Indonesian nationalism. This locus does not have a geographical context, rather it is a geocultural one. If, geographically, the Indonesian nationalism is located in the archipelago of the Dutch East Indies, then in this cross-region national constellation, visualizations of the Indonesian nature in paintings become the imaginary space to develop a national consciousness as "Indonesians". In other words, the Indonesian nationalism imagined itself like the beautiful landscape depicted in Mooi Indie ("Beautiful Indies") paintings from that period.

The analogy of Mooi Indie paintings refers to the development of orientalist social theories during the late 19th century until early 20th century. The rise of these theories were directly related to the Western colonialism and colonization over nations outside Europe, particularly in Asia and Africa. The theories offered a new ideology that inspired the dawn and development of nationalism in the Dutch East Indies region.[iii] By then, the practice of visualization through art and photography was already established and enhanced the process of constructing the narrative. This setting of beautiful natural landscape became a representation of Indonesia in the visual context. Such visualization placed the Indonesian national identity inside an exotic geocultural frame - both appealing and alien. This setting of beautiful natural landscape also represented the extensive geographical proportion of Indonesia. In this perspective, it is easy to overlook the reality of natural landscapes in Indonesia, mostly left out of the painting and therefore unrepresented by the beautiful setting. This situation grows problematic when the visualization of nature is paired with the narrative of natural wealth. Our imagination about the vast beauty of nature brings us to the narrative of abundant natural resources in Indonesia.

Enveloped in the spirit of nationalism, the narrative of “a vastly beautiful and rich nature” seems necessary to fuel confidence for the nation. If the Indonesian nationalism is indeed built through a process of collective imagination as stated by Benedict Anderson (1983), then this imagination needs the right setting. Imagine the narrative to operate like this: the beautiful nature represents majesty, the vast nature stands for greatness and grandeur, and the natural wealth presents freedom. This operation portrays a national image about power, as a large, honorable nation endowed with an abundance of natural resources. In this context, the state needs images of beautiful nature to shape the narrative of looking at Indonesia, turning it into a political project. On one hand, this political project is about freedom, where the rich soil can be harnessed to attain self-sufficiency and prosperity. On the other hand, since the natural resources are controlled by the state, it also demonstrates how the state sees their power. Ultimately, the beautiful nature is a projection of socio-political and socio-economic stability. Nevertheless this projection is considerably detached from reality of nature, the actual wilderness that is elusive and unpredictable and at the most extreme, can grow out of control.

The dream of Indonesian nationalism is none other than the dream of independence, as stated in the preamble and the articles of the 1945 Constitution. When this dream of independence is projected into operating the aforementioned nationalist imagination, then what we get is the image of a wealthy nation due to the abundance of resources. Recognizing nature as a resource is an example of how nature is quantified economically as one production factor. Nature is seen as a source of raw materials for production that shall fulfill the livelihood and necessities of the people. This objective to ensure a fair and prosperous society as stated in the preamble of the Constitution comes across as a justification for extractive practices on nature. This principle gives the impression that when it comes to nature, the Indonesian nationalist dream is binary. This impression gives way to an assumption that nature can potentially threaten self-sufficiency and prosperity - especially when self-sufficiency and prosperity are only acknowledged in a capitalistic economic context. Before we realize it, nature is put as a threat to nationalism. Stability, both socio-political and socio-economic, is the important keyword played by the state during the New Order. That was why the regime adopted the “environmental” approach to nature as an effort to inject “control” to the Indonesian soil. Although this approach pairs the wilderness and the people’s living sphere, at the same time it also demonstrates the state’s attempt to control the environment, which is to turn nature into a source of raw materials that can be extracted.



Culturally, the New Order placed nature inside a frame of tradition. Furthermore, the regime put tradition as the backdrop for the tourism industry. If we are to critically assess, we can comment that the New Order used the tourism industry as an instrument to convert the untamed wilderness into organized and controlled beautiful landscapes. Nature was no longer a place of savagery, but an orderly environment. This situation built yet another narrative about food security, which widened the gap between “nature” as environment and nature as wilderness or forests. Forests was the place for everything uncontrollable and wild, so forests needed to be cultivated and turned into a resource. If not, then because of its wilderness, forests would lose in the battle against the changing time. There was no other choice, forests had to be conquered (read: cultivated).

In this situation, we can say that the New Order has pulled forests off the narrative of nature and in the cultural context, they have removed forests from tradition. Then what about villages and rural areas? Indonesia is an agrarian nation who has lived from agriculture for generations. This is a prevalent narrative taught in primary schools through the dichotomy between “village” and “city”. In Indonesian Language classes, “spending the holiday at grandma’s house in the village” is a classic theme in many reading materials and writing assignments. The grandparents live in the village, so are the farmers. So, it can be inferred from the narrative that our grandparents and the farmers are of the same status. This position symbolizes tradition, but moreover it indirectly transmits a message about being trapped in the past, far from modern civilization or to take it to the extreme, “lowbrow peasants”. This is perplexing, because how can we exercise the strategy for food security in the future if we consider the forefront of this development as those who are backward and behind time?

The food security program during the New Order glorified farmers on the national television channel TVRI[iv] and at the same time framed them as traditional figures who could not adapt to the changing time. In my personal experience, I remember that since I occupied myself with stereo-set, playing vinyls and cassettes became my favorite after-school activity, the images of farmers and villages that I learnt in primary school slowly withered from my mind. Agriculture seemed dull. Rice fields, paddies, and cows became a narrative that I resisted because it was the antithesis of my imaginations of the new world, a world I discovered through popular culture. Listening to electric, earsplitting sounds of Rush while I flip through the pages of sci-fi novels by Victor Appleton or Isaac Asimov opened my eyes to a new narrative about the future. Star Wars offered a concrete image of what the world would look like in the future - a world driven by the narrative of technology, inter-galactic adventures, and dynamics of the universes. Since then, I became obsessed with the future:

a bright and challenging ultra-modern world. Not once I thought about becoming a successful, rich, or famous person. All I wanted was to be in that inter-galactic future, that alone would suffice. However, fast forward to today, I cannot find that future in the contemporary daily lives. Instead, in the similar narrative, I have discovered how the future of myself and the entire humanity is incredibly frightening. If “the entire humanity” is too much to comprehend, let’s focus on Indonesia.

Perhaps the fundamental problem of the disappearing narrative about land, water, and conservation is because this topic is often placed at an intersection between modernity and tradition, progress and backwardness, fulfilling livelihood of the masses and sustainability of the natural ecosystem itself. Nature is represented as the untamed “forests”, while in the discourse of Indonesian development forests are placed in the cultural geography of “villages”. Villages become a geopolitical unit expected to serve the needs of cities. In this setting, villages gradually transform into new cities when their residents grow distant from nature and see the forests for their economic purpose as a resource to be extracted. In a swiftly changing civilization, nature and forests in this context are merely economic variables that may have to perish one day. But we cannot let nature perish along with the disappearance of forests from the narrative of human civilization. Professions like postmen and graphic designers can be replaced completely[v] with the changing technology, but what can we do once nature is gone?

Today, our daily lives are stuffed with conversations around the economy and politics to how much internet quota we have left. We no longer mark the changing seasons through the movements of the moon and the sun, but instead from the information about discounts at the shopping malls or the notifications from online shop apps on our mobile phones. We can be informed of our heart rate just as we observe today’s and tomorrow’s weather, we can even gather all the latest updates from earthquakes around the world without having to move ourselves as everything is accessible on our fingertips. The mechanical technology is steadily replaced by digital technology that offers much faster access to information. At this point, it is hard for us to realize how technology - essentially the brainchild of science - has seized almost all of our everyday life.  We can fall into the assumption that the present of technology is only natural. Especially when the technology comes in the form of access to information.  Information - also part of science - creates an illusion of a new reality. This new reality becomes the basis for deciding policies and making important decisions. The fluctuations of global oil prices can affect the price of goods sold in grocery stores in rural areas in Jakarta. This kind of situation makes the illusions of the new reality seem natural.



The critical issue is actually the absence of alternative narrative, or if necessary, a resistance to the dominant narrative. If so, then the imagination of technology can dominate the entire human conversation about themselves and their relations with reality. The dominant discourse of technology can possibly fuse images of the future of humanity and abandons the rich cultural narratives. Slowly, our daily life has immersed itself into situations described in sci-fi Hollywood movies about the future. We are so used to seeing the future state of humankind in apocalyptic settings in sci-fi movies that we consider the threat of environmental destruction is merely fabricated for entertainment purposes.  Furthermore, we cannot imagine the future without advanced technology, as if the future of humanities is only about technology as the paramount achievement of human civilization. Technology has replaced our natural reality so much that we are accustomed to think about how technology can solve all human problems, including the problems with nature. If we can successfully conceive an imaginary future society who lives in a new world order named Indonesia, can we also imagine the kind environment they live in?

4. Unity in Diversity in A Pot

*The Commons*, by artist collective Digital Nativ, at first glance seems like a giant pot filled with various plants. But, *The Commons* is not just an ordinary pot - the plants inside comes from Katingan conservation forest[v], which laid on peatlands[vi] on the banks of Mentaya river,[vii] that crosses East Kotawaringin Regency, Central Kalimantan. The conservation forest covers an area of approximately 150,000 hectares and is a Carbon Finance project[viii] by PT. Rimba Makmur Utama (RMU), a company based in Bogor, West Java.[ix] Supported by the Katingan-Mentaya Project, Digital Nativ brought the various plants and arranged them in the pot, making this work as a miniature or simulation of the peatland forest ecosystem in Katingan. With this artwork, they also produce a narrative about the importance of ecological resilience. This is conveyed by Digital Nativ through the installment of water spray, light along the circumference of the pot, and a lamp on top of the pot that serves as artificial sunlight. The water comes from a tank inside the pot to display simulated peatlands, while the beaming light from the pot is actually created by electric signals of the plants that are converted into light.

This technical strategy is enabled because in 2017, Digital Nativ together with a British new media studio Invisible Flock has developed a simple technology to seize electrical signals produced by plants. This simple technology is an art project called “Nada Bumi”[x], or Earth Tones. In short, Nada Bumi enables us to directly see/hear/feel the process of photosynthesis and electric signals emitted by plants. In June 2019, during the research and experiment stage for *The Commons*, Digital Nativ brought the Nada Bumi device and tested it directly to all plants and vegetations in the Katingan forest area. The device was connected to a MIDI Tone Generator that transformed electric signals seized by Nada Bumi into audio signals. The result amazed many conservation workers there. This explains that the "Sound of Nature" is not just a symbolic expression, but also a real scientific experience[xi] In a symbolic sense, the "Sound of Nature" represents a divine dimension of nature. It presents nature as blessing and an undisputable embodiment of God. To be able to listen to these sounds, one will need a specific ability, such as a metaphysical ability far from the discourse of natural science.

Since the validity of such metaphysical ability is scientifically debatable, as a result the "Sound of Nature" can only be approached in the context of elusive symbolisms. The breezing wind, the rippling wave, the blossoming flowers, the morning dew, and similar natural phenomena are assumed to have an audio quality denoting beauty. The Nada Bumi device invented by Digital Nativ can be taken as a rebuttal to this symbolic approach. In *The Commons*, the concept of Nada Bumi is applied slightly differently. It is not the "Sound of Nature" that they display, but instead it is the beaming light that serves as a metaphor for "the light of nature". The metaphor can be interpreted by some who tends to appreciate the reality of nature as a visual phenomenon that is easy on the eyes and unproblematic.This is exactly where the previously mentioned problem of "natural scenery" lies, when nature is perceived through an incontestable divine perspective, yet at the same time we treat it as an object to be extracted endlessly. The images of beautiful natural landscapes fail to teach us how to properly utilize the rich biodiversity in Indonesia.

As an example, the biodiversity in peatlands of Katingan. At 150,000 hectares, the conservation forest is only a part of the entire 2.7 million hectares of peatlands in Central Kalimantan. Indonesia has the second largest peatlands in the world, with an area of 22.5 million hectares.[xii] The peatlands function to absorb 30% of the world's carbon, prevent droughts, and impede the contamination of saline water to agricultural lands. This means that if Katingan forest alone can store almost 34 million tons of CO2, then how much potential contribution that can be provided by the peatlands in Kalimantan in responding to the challenge of climate change? *The Commons* reminds us that in addition to taking actions on climate change, the Indonesian biodiversity (in this case the peatlands) offers much more than just a beautiful scenery and economic growth: life. It is obviously relevant that climate change is a threat to mankind. According to biotechnology expert Nico Wanandy,[xiii] peat forests can achieve their optimum state in a balanced environment, where biodiversity is preserved. The practice of



monoculture - cultivation of a single crop in a given area - is potentially harmful to the characteristics of the soil. When the soil becomes weak and unable to support life, it disrupts the balance of the ecosystem. Indeed, ecosystem balance can only be achieved through natural biodiversity.[xiv]

The emphasis on Wanandy's statement is what makes peatlands rich is the untamed biodiversity. This untamed wilderness is a direct contrast to the "beautiful" nature controlled by humans. The beautiful nature here represents humans' desire to control nature through scientific thinking and is biased by economic interest. This bias happens because humans view nature as source and raw materials for endless (economic) production, in the similar perspective nature is also seen as a complementary component for industry. That being said, untamed wilderness is not a character that can be tolerated by industrial economic practice, where everything must be carefully planned and calculated. Therefore, strict control mechanisms are of utmost necessity and any characteristic of wilderness must be conquered. Including nature. In this position, science is often used as an instrument to gain that power.

Images of beautiful nature become an instrument to advance the hegemonic narratives of nature. Firstly, the hegemony of nature as an infinite resource that can be exploited entirely for the prosperity of mankind. Secondly, the hegemony of beautiful nature as an indicator of the abundantly rich Indonesia. This narrative creates the illusion that since humans can do anything to nature for the sake of prosperity, then nature is always under the control of humans. Nature is therefore not wild, but beautiful and orderly or in other words safe and under control. This hegemony enables the state to approach nature through the political lense, that aims at safety and stability. Thirdly, the image of beautiful Indonesian nature that extends from Sabang to Merauke represents the geopolitical stance of Indonesia. Such images appear to embrace Indonesia in a single visual representation. However even this representation is a controlled representation that leaves no room for different articulations and expressions. The power over the wild characteristics of nature is similarly appropriated into the control over the variety of articulations and expressions. This illusion that nature can be altogether overpowered confirms the assumption that societal dynamics, too, are under control. Because after all, don't we humans live in harmony with nature?

The presence of *The Commons* inside the exhibition room of the National Gallery of Indonesia is just one articulation and expression about the reality of Indonesian nature, that s rich and diverse. This work rebuts the assumption that there is only one trope of beautiful Indonesian nature. Because, in reality every place across the archipelago has their own distinct characteristics, which requires different ways of management. On one hand, this artwork also underlines the importance of considering different ways of seeing nature, outside the academic tradition of natural sciences. We cannot simply assume that any non-scientific narrative about nature is a myth or rumors. What we can do, as initiated by this artwork, is to try to perceive nature using multidisciplinary approaches. This is not to say that science is not relevant, it is, but so is cultural approach that can be a multi-faceted instrument for science to interpret and appreciate biodiversity. Perhaps *The Commons* is indeed repeating what has been done by traditional communities for thousands of years: to have the wisdom to acknowledge that we are visitors in nature, not the other way around. However, this reiteration must be recognized as a contemporary strategy to understand how humans can live harmoniously with nature.

**Further reading:**

i Droste and Gossel, eds. 2005. Bauhaus

ii Zaelani, Rizki A. 1999. “Landscape Painting”, from the curatorial introduction for the exhibition Indonesian Modernity in the Representation of Fine Arts. National Gallery, Jakarta, May 8 - June 8, 1999, page 47. Source: “Reka Alam: Praktik Seni Visual dan Isu Lingkungan di Indonesia (Dari Mooi Indie Hingga Reformasi)” - IVAA Data Catalog Series #2.

iii Onghokham. “Freezing Indies” in Bachtiar, Carey, Onghokham, eds. 2009. “Raden Saleh, Anak Belanda, Mooi Indie dan Nasionalisme,” Depok: Komunitas Bambu, page 173 - 180. Source: “Reka Alam: Praktik Seni Visual dan Isu Lingkungan di Indonesia (Dari Mooi Indie Hingga Reformasi)” - IVAA Data Catalog Series #2, page 30.

iv TVRI or Television of the Republic of Indonesia was the only national television in Indonesia from 1962 to 1989. The monopoly ended when Indonesian private television appeared in 1989 (Author)

v The conservation forest is managed by the Katingan Mentaya Project, a conservation project managed by PT. Rimba Makmur Utama, a company engaged in Carbon Finance. More information about Katingan Mentaya Project can be found in the following link: https://katinganproject.com

vi According to katadata.co.id, "Peat is a wetland rich in organic material formed by accumulations of decayed organic materials for thousands of years. Its existence has various benefits, among other things, peat can store 30% of the world's carbon, curb droughts, and prevent contamination of salt water into agricultural irrigation. In addition, peat is also home to endangered species. In the article “Indonesia Has The Second Largest Peat Area in the World”, based on Global Wetlands data accessed on April 16, 2019, Indonesia has the second largest peatland in the world with an area of 22.5 million hectares.
https://katadata.co.id/infografik/2019/04/29/luas-gambut-indonesia-terbesar-kedua-di-dunia, last accessed on 2 April 2020.

vii For more information on Mentaya River, please access http://www.kotimkab.go.id, and https://id.wikipedia.org/wiki/Sungai_Mentaya

viii Carbon Finance (CF) is a generic term used for the revenue streams that can be generated by low-carbon projects and activities from sale of their global greenhouse gas (GHG) emission reductions by sources, or GHG emission removals by sinks, or from trading in carbon credits. It has been one of the World Bank Group’s first and longest engagements for mitigating climate change.
Source: https://ieg.worldbankgroup.org/evaluations/carbon-finance.



ix Through this conservation forest management, RMU states that it is able to prevent the release of greenhouse gases equivalent to 33,965,359 tons of carbon dioxide (CO2). Source: https://katinganproject.com website accessed on April 1, 2020.

x For more information on Nada Bumi, please access
http://invisibleflock.com/portfolio/nada-bumi/ and
https://www.britishcouncil.id/uk-indonesia-2016-18/cerita-kami/wawancara-invisible-flock

xi Using this device, one can hear the difference between plants that receive enough water and sunlight, because of the direct response from the plants (that changes its tonal character) when someone performs an action to it. Once, Miebi Sikoki (a member of Digital Nativ) placed a Nada Bumi device on a plant inside a pot at the central dining room of RMU in Katingan. The plant was fern, planted on a peat soil taken from the forest. That indicated that fern had healthy soil and enough water, not to mention placing the pot outside the dining room also allowed it enough supply of sunlight. The sounds produced from the plant then were a smooth, soft dynamic. However, when the pot was moved inside the dining room and thus deprived it from direct sunlight, the sound changed into a messy, harsh dynamic (Author)

xii According to this source, Indonesia has the second largest peatland in the world, after Brazil. https://katadata.co.id/infografik/2019/04/29/luas-gambut-indonesia-terbesar-kedua-di-dunia.

xiii Nico Wanandy is an expert in biotechnology and molecular biology from the University of New South Wales (UNSW) Sydney at the School of Biotechnology and Biomolecular Science. Wanandy is also a consultant for soil improvement at URDI (Urban and Regional Development Institute) and PT. RMU (Rimba Makmur Utama). Currently Wanandy focuses his research on native Indonesian farming practices, which he believes to be very wise and mindful in harnessing nature (Author).
For additional information about Wanandy, see the link
http://urdilearningforum.urdi.org/nico-wanandy/

xiv Based on an interview with Nico Wanandy, Ph.D at the conservation camp of Katingan Mentaya Project by PT. RMU in Katingan, East Kotawaringin Regency, 19 June 2019 (Author).

# CHAPTER II
# SUSTAINABILITY

## *Curatorial Essay*

### Sustainability and Media Arts
**Irene Agrivina**

Artists working with new media often work alongside scientific researchers to provide different approaches and raise public awareness toward environmental issues. Topics such as pollution and contamination in our aquatic ecosystem, global warming impacts to weather patterns, orbital debris in space, gardens in the sky, green design alternatives, and resource consumption put forward new discourse on the urgency and importance of sustainability. Due to its specificity, artists working on a particular subject can be further classified as biogeneticians, space artists, industrial designers - but everyone is part of the umbrella term “environmental artists working with new media”. By producing artworks that are relevant to current ecological problems, these artists open a space for dialogue to search for future possibilities and solutions.

# Eva Bubla

ARTIST STATEMENT

The gradual disappearance of rural and wild areas, and their transformation into densely populated cities and urban zones are leading to a series of sustainability problems. The often debatable city planning, the elimination of green areas, or heavy traffic all contribute to the degradation of urban air quality. Regulation No. 26/2007 of the city of Yogyakarta claims that 30% of the total area of the city must be used as green open space (Ruang Terbuka Hijau RTH), from which 20% is public RTH and 10% is private RTH. These areas could play a significant role in air pollution abatement, however, case studies show that this number is not yet fully implemented in practice. Furthermore, air pollution is not only an outdoor problem, but an indoor problem as well. Household activities, like cooking, emit various particles that are harmful for the environment and human health, while smoking further degrades the quality of indoor air. The project aims to raise awareness on the effect of these human activities as well as the importance of green areas and plants in tackling pollution levels. Inspired by a 1989 NASA Report which researched and tested plants that are the most effective in purifying indoor air, the project researches and uses the Sansevieria plant (often called as snake plant or lidah mertua) to critically reflect on the topic of air pollution. The installation invites the audience to inhale the air ventilated from an incubated plant modul.



Designated Breathing Zone
Glass box, Sansevieria plant, micro-controller, and LCD TV
Variable dimension
2019

# CHAPTER III ARTIFICIAL INTELLIGENCE

## *Curatorial Essay*

### AI and Art in the Post-human Era
### Jeong Ok Jeon

Numerous efforts are being made to understand human beings in different ways. Under the umbrella term of "post-human,"[i] many schools of thoughts are now trying to redefine the notion of human in the 21st century because it is believed that the notion of human in the traditional sense[ii] that came along with modernism does not reflect who we are in the new millennia. With the rapid development of cutting-edge technologies such as biotechnology, artificial intelligence, the Internet of Things, and virtual reality, we are witnessing that the nature of human and its relation to the world has dramatically expanded. Human body and mind is enhanced by technology[iii] while intelligent machine replaces man in various work places; humans are infinitely connected with things while freely traversing the digital world, which shifts the perception of time and space. In such circumstances, human and/or humanity is challenged and transformed fundamentally.

Challenges toward faith in human and humanity is the central debate in this age of post-human. Artificial intelligence, one of the major technological advancements in the new age, proposes a new challenge toward the faith in human intelligence. A 2016 Asian board game match between a computer and a human player marked a symbolic event in the history of human intelligence. AlphaGo, a computer algorithm developed by DeepMind, a Google's artificial intelligence company, defeated Korea's "Go" world champion Lee Sedol with a score of 4-1. According to DeepMind, AlphaGo's algorithm is at the point where it does not only master a series of games itself, but also thinks more intuitively[iv]. A computer algorithm is now able to assimilate one of our unique human characteritics: intuition. Each year, AlphaGo's evolution was groundbreaking that it became more and more autonomous. It enhanced itself through reinforcement learning as found in AlphaGo Zero, which learned from other machine rather than by human supervision[v]. Now a self-taught algorithm, AlphaZero can master various challenging rules of, not just the game of Go, but that of chess and shogi[vi].

Without notice, AI technology is available in our various day-to-day activities and tasks, and it can no longer be separated from our lives. Every time of our commute, we automatically rely on GPS navigator in order to find the best way of getting to our destination. GPS navigator is getting smarter and more accurate that it keeps updating itself in real-time in order to provide us with the best route option. Meantime, unlike GPS traffic information that allows us to make our own choice, other fields that incorporate machine intelligence can make a decision that we must obey. The Atlantic League is the first American baseball league that let a computer make a judgement on balls and strikes[vii] and US Major League Baseball is now considering changing from human umpires to an artifically intelligent ball-tracking system[viii]. We will soon witness that the AI will determine our destiny and give us a direction for what must be done.

To many, the aforementioned scenarios such as human-machine tournament (AlphaGo), human-machine collaboration (GPS), and human-machine command system (Baseball umpire) might be a wake up call to the possible shifts of our current and future upper-hand position as human beings. In other words, it could be a good opportunity to acknowledge the fact that human is no longer the center of intelligence; moreover, biological human might have to prepare for a new society where all kinds of life forms live happily and peacefully together. As much as natural humans are merging with machine, and artificially manufactured machine is being assimilated to natural humans, the boundary between human and machine becomes blurred. Therefore, various fields of disciplines attempt to explore the rapid changes of the current situation occurred to human and to prospect what may come in the next with a post-human perspective.

Five Passages to the Future is an exhibition that aims to explore the rapidly changing phenomena in the age of post-human and how science and technology has influenced contemporary art and artist. As one of the five chosen themes (eco-politics, sustainability, AI, digital narrative, wearable technology), 'AI' exhibition features a Korean new media artist Cho Eun Woo who has long been contemplating the relationship between human and machine. Through her interactive new media art installations, Cho examines how convergence between art, science and technology can be carried out and what kind of new aesthetics is possible in the age of post-human. Using the latest technology of artificial intelligence and robot engineering, Cho created new body of works for this exhibition; *'AI, Brainwave and Hyper Protocol City'* (2019) and *'Robot, Cyborg and Human'* (2019).

*'AI, Brainwave and Hyper Protocol City'* (2019) is a continuing project of *'AI, Brainwave, Ideal City'* that she has been developing since 2017. The installation series consists of a group of geometric structure that resembles an architectural colonnade in ancient city, which is imbedded with colorful LED lights for the futuristic effect. When audience wear a remotely controlled brainwave device and concentrate their mind in front of the work, the lights in the installation become illuminated as if it reflects the hyper network society where things are manipulated by the performance of human and artificial brain. On the other hand, for *'Robot, Cyborg and Human'* (2019) Cho used a multifunctional robot module called Pingpong, produced by a Korean robot company Robo Risen where she serves as the Chief Officer of Branding and Design. The installation presents four hybrid humanoid figures that are controlled by a Pingpong-based remote gun. When audience virtually shoot the gun toward the humanoids, the humanoids spin around, and then rip off the artificial human skin attached to the gallery wall, which results in the symbolic human blood dripping to the ground. By targeting each other, this installation questions whether it is human or humanoid that asks who the true human being is.



Cho focused on the two sides of technology – positive and negative - and tried to express the ideal world where human and AI coexist in accordance with their purpose. With her works, Cho helps us to experience correlation between human existence and science technology. Robots and cyborgs are constantly evolving and unveiling new possibilities for humanity. By bridging distinct ideas between technology on one hand and nature on the other, the artist demonstrates the capabilities of artistic imagination that scrutinizes the most urgent human inquiry for the future. Furthermore, facing the era of Industry 4.0 and entering the new age of post-human, it is intriguing to explore new aesthetics of new media art that incorporates AI technology. It seems inevitable for us to embrace machine-integrated lives, but it is crucial for us to be conscious of, and keep in check, where our machinized world will lead us. Following questions shall help us for self-reflection: what is human being?; how has its concept been challenged?; what kinds of questions AI asks to human beings, and vice versa?; and how are art and culture influenced by the new relationship between human and robots/cyborgs if more and more of the kind species live together with humans?

**Further reading:**

i According to Francesca Ferrando, there are various schools of thoughts that address the topic of post-human in different ways including, but not limited to Transhumanism(s), Posthumanism(s), Antihumanism(s), New Materialism, and Metahumanism, and they are plural because each of them is developed by various sub-movements such as Democratic Transhumanism, Libertarian Transhumanism, Singularity, Critical Posthumanism, Cultural Posthumanism, Philosophical Posthumanism, and so on. Influenced by Rosi Braidotti, Ferrando also stated that the notion of human is an open frame in the sense that it can be understood through evolution (Darwin), technology (cyborg), and ecology (anthropocene). Ferrando, Francesca. What does “posthuman” mean? (2017). http://www.theposthuman.org/course-the-posthuman.html. Last accessed on January 30, 2020.

ii Traditionally, human beings can be defined as (1) human animal that possesses a life; (2) human rationale that possesses an ability to speak and calculate; and (3) human autonomous that possesses free will, responsibility, and dignity. Paek, Chong-Hyon. Confused Concepts of Human Beings and Problems of Post-Humanism (2015: pp.128-140). http://dx.doi.org/10.15750/chss.58.201511.005. Last accessed on January 29, 2020.

iii A simple example of body enhancement is to use a prosthesis such as a pacemaker while an example of mind enhancement is ‘mind uploading,’ the process of transferring an intellect from a biological brain to a computer. Bostrom, Nick. The Transhumanist FAQ Version 2.1. (2003: p.7 & p.17). http://www.nickbostrom.com. Last accessed on March 16, 2020. Likewise, some transhumanists believe that human life can reach 1,000 years. It is plausible because a human's physical body can be extended by replacing its natural organs with permanent machines so that we eventually become cyborg (cybernetic organism). Paek, Chong-Hyon. Confused Concepts of Human Beings and Problems of Post-Humanism (2015: p.146). http://dx.doi.org/10.15750/chss.58.201511.005. Last accessed on January 29, 2020. Accepting the speculative idea of ‘mind uploading,’ to lengthen the human lifespan is also plausible because a human's mind can be uploaded into a computer and transferred to other physical forms which may result in many of the same people living alternatively.

iv https://deepmind.com/research/case-studies/alphago-the-story-so-far. Last accessed on January 27, 2020.

v Silver, D., Schrittwieser, J., Simonyan, K. et al. Mastering the game of Go without human knowledge. Nature 550 (2017: pp. 354–359). https://www.nature.com/articles/nature24270.epdf?author_access_token=VJXbVjaSHxFoctQQ4p2k4tRgN0jAjWel9jnR3ZoTv0PVW4gB86EEpGqTRDtpIz-2rmo8-KG06gqVobU5NSCFeHILHcVFUeMsbvwS-lxjqQGg98faovwjxeTUgZAUMnRQ. Last accessed on January 27, 2020.

vi Silver, D., Hubert, T., Schrittwieser, J. et al. A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play. Science Vol. 362, Issue 6419 (2018: pp. 1140-1144). https://science.sciencemag.org/content/362/6419/1140. Last accessed on January 27, 2020.

vii Bogage, Jacob. Baseball’s robot umpires are here. And you might not even notice the difference. The Washington Post. July 11, 2019. https://www.washingtonpost.com/sports/2019/07/10/baseballs-robot-umpires-are- here-you-might-not-even-notice-difference/. Last accessed on January 27, 2020.

viii Bakalar, Jeff. Meet the AI that could replace your baseball umpire. CNet. August 13, 2019. https://www.cnet.com/news/meet-the-trackman-ai-that-might-replace-your-baseball-umpire/. Last accessed on January 27, 2020.





# Cho Eun Woo

Inspired by the Italian Renaissance painting The Ideal City (1480–84) attributed to the architect and artist Fra Carnevale (1420–1484), in her installation AI, Brainwave & Hyper Protocol City, the artist created a diorama of spectrum reflective plexiglass panels and columns placed in a grid filled with thousands of LED bulbs connected to sensors responding to a brain wave program. The program takes input from a wireless headset responding to EEG signals from the brain of the participant, which in turn activates the lights within the columns.

The second work in the show, Robot, Cyborg & Human, is an installation consisting of a series of life-size human figure cast in golden polymer. Each human figure contains a tiny robot (supported by Pingpong, Robo Risen) inside. Depending on the input from the main robot’s signal, the four human figures emanate a multitude of action and movement.

These installations function in a similar way – in the sense that there is an interaction within the process of the viewer, the robot, and the artwork – that interfuses the whole scenario into a complete circuit. The work is bridging the distinction between science and cultural meaning – posing as a visual metaphor that brings science into the emotional domain of public discourse that opens up the process of self-reflection. We can imagine alternative possibilities, which bring us into a state of awareness, reformatting fragments of stimulation into alternative realities and new narratives.


CARA MENGENDALIKAN 4 ROBOT
HOW TO CONTROL THE 4 ROBOTS


Ai, Brainwave & Hyper Protocol City
LED light, brainwave device, brainwave configure program, multimedia science installation
Variable dimension
2019

Robot, Cyborg & Human
Pingpong robot cube module, multimedia science installation
70cm x 70cm x 100cm each (4 pieces)
2019



## *Critical Essay*

### The Fear, Mind and Reflection
### Aprina Murwanti

In social settings, the Artificial Intelligence (AI) help humanity in technical aspects, for example the prevalent use of face recognition technology in finding missing children and reducing the numbers of human trafficking in India, or how a chatbot[i] can help predict mental illness in the United States. In the art scene, how can AI contribute to the dialectic and extend the emotional aspect of being human? According to renowned contemporary artist Sougwen Chung, AI offers a new way of seeing, self-reflection and opens an opportunity to explore intersubjectivity between human and machine[ii]. However, how can the general public participate and become aware of the important relationship between human and machine? Cho Eun Woo attended to this awareness in her two installations presented for *Five Passages to The Future* exhibition at the National Gallery of Indonesia in 2019.

Along with the development of AI systems, AI is often positioned as a competitor to humans rather than their complement. This contradiction has gradually increased excessive fear that one day AI will take over the roles of humans in the world. The success of AI-generated painting sold at Christie's for $432,500 in October 2018 sent shock waves through the art world, raising speculations about the relationship between art and AI. In particular, it verified the prediction that the existence of AI could give way to a new artistic genre or in some ways ignite a movement that might transform conventional values and boundaries. Understandably, the growing fear disrupts humans' confidence in their own authority and control in the world they live in. With her installations, Cho responds to this inherent human desire to think and control and addresses the collective fear that we may or may not openly admit.

Philosophy is the root of Cho's thoughts on the future of a harmonious relationship between human and machine. Cho's installation, '*AI, Brainwave and Hyper Protocol City*' (2019) displayed in this exhibition is a continuation of '*AI, Brainwave, Ideal City*', a series that she has been working on since 2017. This work takes inspiration from '*The Ideal City*' (1480-84), a Renaissance painting attributed to artist and architect Fra Carnevale (1420 - 1484). Carnevale worked in a time when social and cultural changes brought a new privilege: questioning the norms. This Renaissance humanism fueled radical transformations across Europe during the 15th and 16th centuries. In '*The Ideal City*', originally commissioned by Federico da Montefeltro for his palace in Urbino, Carnevale proposed his thinking for an ideal society. The ideal city is a calm, spacious landscape where buildings are placed in such a balance and harmonious composition; for Carnevale, this is the ideal city planning for virtuous rulers, who seek prosperity and welfare for their people. While Cho does not display a city landscape the way Carnevale did, she utilizes Carnevale's ideas of a city as a metaphor, a symbol for an ideal interrelation between humans and technology.

A spacious, sterile and balanced composition can be felt in Cho's installation through lines and symmetrical shapes. Like its predecessor in the series, '*AI, Brainwave and Hyper Protocol City*' demands human participation to activate the installation. There is a device attached as part of the work, which is built with a sensor that can detect brainwaves when this device is put on the audience's head. The use of brainwave detector device makes this installation very interactive. Brainwaves are produced through synchronization of electric vibrations happening in the brains so when the audience puts on the device and concentrates, the device can detect the brainwaves and change the color of the LED lights in the installation. The harder the audience concentrates on critical and creative subjects, the brighter the LED lights will shine and the more dynamic it will move. The colorful lights produced and controlled by the audience's mind are reflected on ten shiny bars of the installation. This bright, dynamic colorful light is an extreme contrast to its rigid, balanced state when there is no one putting on the detector device and activating the installation. Using technology, Cho amplifies the importance of the mind power and highlights the mutual relationship between humans and technology.

Without actual people activating the installation, this artwork seems cold and lacking emotion. Audience may interpret it merely as a glamorous, still object. Indeed, the installation stands firmly with a static composition, this intended appearance highlights form and balance. The use of shiny materials that bounce and reflect lights is a signature style of the '*AI, Brainwave, Ideal City*' series, appropriated to heighten the interactive human minds and thoughts. By turning human thoughts into real visuals of LED reflections, Cho blurs the barrier between human and object. When we activate the installation, we will realize that our presence is fundamental, our thinking is evermore important. In addition, Cho feeds human desire for control and power over technology. The flashing and flickering light reflections of an activated installation serve to glorify the power of our thoughts. We are made to feel powerful and grateful to see how our own brains work in such a magical manner. Once we take off the detector device and see the lights going dim, we will miss the magical momentum. Perhaps then we reflect and recognize how we need technology to address our needs and amplify our ego. Strangely, this experience forces us to upgrade our level of consciousness and at the same time accrue our respect for technology.



The human brains and AI have different ways of working and functioning. According to philosopher Andy Clark, human perceives the world by information that came their bodies, brains and the environment[iii]. It is only when the external environment, bodies and brain get together, that humans can experience the world. Thus, in order to understand the world, something – including AI – has to correspond to actions and perceptions. Clark emphasizes that even if we humans can do amazing things, we will not really understand the world if we cannot access the causality in the world. Human brains can work on themselves and synchronize information, which AI cannot achieve perfectly. In order to develop a general intelligence, AI system has to have the capability to embody interactions with the world. For now, AI is limited to a passive, non-embodied entity with special purpose or singular function. AI will be super powerful if and when they are able to "talk" with each other, like human beings. The human experiences, attitudes, habits of thinking and perceiving are inseparable from the constant process of the brains to rewire and improve themselves. Therefore we do not need to fear the technology, but what is more concerning is we have to fear ourselves if we give in to the fear of losing our power to machines.

With her installation, Cho Eun Woo explores and proposes coexistence between human and machine in the same habitat, a complementary ecosystem. Cho's interest in artificial intelligence is her way to reflect on her personal life and her hope to contribute to a harmonious relationship between humans and technology in the future. Reflecting on the most critical periods in her life, Cho realizes that humans cannot control the universe through their powerful thoughts alone, however a strong mind can change the self and society.

*"…my installation work leads us to experience dialectically about the correlation between human beings and science and technology, and the world of multiple objects is gathered in a common space and time ('co-being', simultaneous with other beings)" (Cho, 2018).*

Her role as the Chief Officer of Branding and Design at a robot company Robo Risen, makes technology an inseparable part of her artistic approaches and practices where she combines influences from both disciplines instead of just using technology solely as an instrument to communicate her message. Robo Risen supports Cho's second installation in the exhibition, 'Robot, Cyborg and Human' (2019), which uses a Pingpong multifunctional robot cube module produced by the Company. In this artwork, the audience can control the figurative robots using a pistol-shaped controller. According to the testimonials from the gallery sitters at the National Gallery, this installation is among the audiences' favorites, which could be attributed to its interactivity and power-evoking role-play. The audience may feel their sense of authority as they control the movement of the four figurative robots standing on the pedestal. The pistol, a symbol of power and strength, fuels their ego and addresses human-machine engagement.

In this power-hungry society, Cho's installations become addictive and reflect the fragile, greedy humans who seek to fulfill their endless desire to control things and to have the power over every situation, including the future. However, eventually, the future of humanity will be shaped by art and technology[iv]. Thus, instead of getting fearful, what if we build our way toward the future by developing our thinking and support those who push the limits of possibilities?


**Further reading:**

i Pruvost, Maxime (2019), 'Artificial Intelligence can Help Humanity - Towards AI', in Medium, July 10, 2019, https://medium.com/towards-artificial-intelligence/artificial-intelligence-can-help-humanity-574f17684923, accessed 4 November 2019.

ii Rea, Naomi (2018), 'Has Artificial Intelligence Brought Us the Next Great Art Movement? Here Are 9 Pioneering Artists Who Are Exploring AI's Creative Potential', in Artnet, November 6, 2018, https://news.artnet.com/market/9-artists-artificial-intelligence-1384207, accessed 5 November 2019.

iii Segal, Michael (2019), 'A More Human Approach to Artificial Intelligence' in Nature, July 24, 2019, https://www.nature.com/articles/d41586-019-02213-3, accessed 10 November 2019.

iv Howard, Lindsay (2017). 'Inventing the Future: Art and Technology' in Invention, ART21 Magazine, October 26, 2017, http://magazine.art21.org/2017/10/26/inventing-the-future--art-and-technology/#.XdrFoC2B3GI, accessed 15 November 2019

# CHAPTER IV DIGITAL NARATIVE

## *Curatorial Essay*

### Virtual Reality and Human Perception: a story in the making
Nin Djani

From religions to political propaganda to modern advertising, narrative plays a pivotal role in human civilization. Historians believe Homo sapiens' ability to create, spread, and believe fictions is what led our species to dominate the world in the last 70,000 years. Narrative – be it fictions and/or factual stories – becomes a technology that helps shape human perception of themselves and the world around them, where a shared belief in certain ideas can form a strong, spiritual bond. For the most part in the Anthropocene, the production of narrative has been reserved to the powerful elites such as kings, priests, political leaders, and public figures. Today, with the advanced technology and ubiquitous media everyone can have their own stories and alternative realities matter. Narrative is now a participatory and interactive experience enabled by the digital world and algorithm.

A relatively recent addition to this digital culture is virtual reality, which extends our real-life sensory capacities into the computer-generated environment. Our senses of sights, hearing, and touch are stimulated to provide us with a synchronized experience of reality. The idea to stimulate the senses is not a new concept per se, since many technological inventions and artistic works aim for such impact. Nevertheless, what differentiates virtual reality from the previous media is the way this technology enables total immersion into the simulated world and becomes a transformative, personalized experience. In the art world, virtual reality allows an entirely new way of exploring, creating, and experiencing art. As the audience is taken into the simulated world to involve themselves as they please, the line that separates the art, the artist and the audience is cut off. Thereby, with virtual reality, each interpretation and every narrative is unique.

*Five Passages to the Future* aims to present what lies ahead for art and humans in this rapidly expanding digital era – the age of post human. As part of the five chosen themes in the exhibition (eco-politics, sustainability, AI, digital narrative, wearable technology), the 'digital narrative' showcase features British artist Shezad Dawood, whose research interweave histories, realities, and symbolisms to deconstruct systems of narrative.

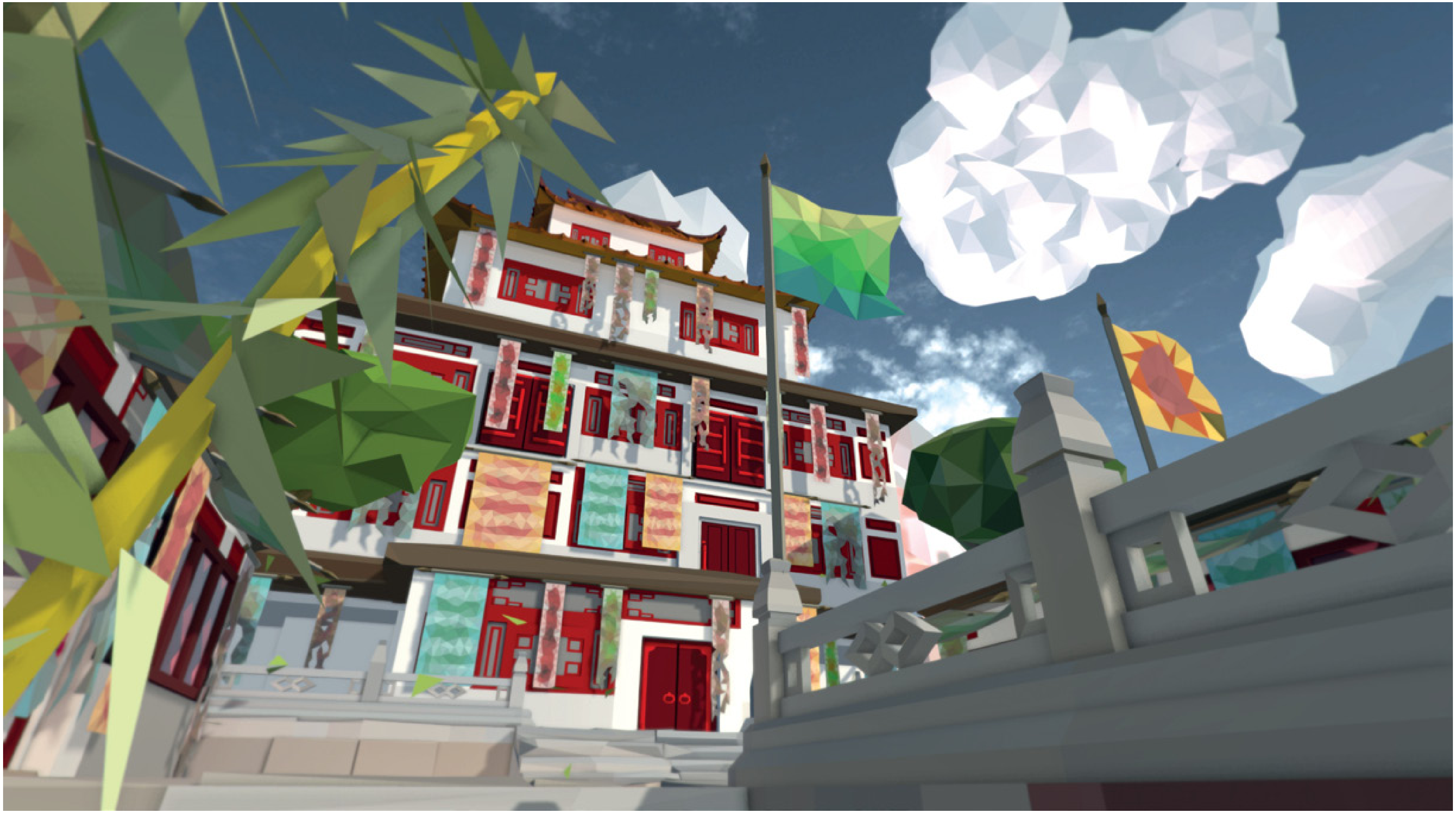
His recent works using virtual reality examines deeper questions of our existence and consciousness. Combining his interests in mysticism and spirituality and his experimental art practice, Dawood’s virtual reality works explore the range of possible intuitive evolution through technology. Set against the backdrop of Cold War South Asia, *Kalimpong* (2016) integrates Tibetan Buddhism philosophy, historical accounts and speculative narratives, inviting the audience to play with their senses, to wander in their imaginations, and ultimately to question reality itself.





# Shezad Dawood

Continuing an interest in how we experience time, and more specifically how the past continues to echo into the present, the artist uses the site of Kalimpong – a small town in West Bengal – as a bridge between the past and the present through an immersive virtual reality. Kalimpong was the site of French explorer and esotericist Alexandra David-Néel's first meeting with the Dalai Lama in 1912, before her legendary journey to Lhasa. Later it was described as a ‘nest of spies’, after the Sino-Soviet split, and in advance of the Sino-Indian war, as various powers competed for resources and influence in Central Asia. Kalimpong also became the base for Texan billionaire Tom Slick's Yeti expeditions, which may or may not have been covert operations. While the works are not specific illustrations of these and other stories, they explore a larger imaginary of place, and the echoes and ripples of the various historical and speculative narrative layers that extend from it.



Kalimpong
Virtual Reality
Variable duration
2016



## *Critical Essay*

### Virtual City, Bardo, Zhuangzi
### Rumi Siddharta

*"The funny thing about games and fictions is that they have a weird way of bleeding into reality."*
*- Erik Davis, TechGnosis*[i]

*"Cities, like dreams, are made of desires and fears."*
*- Italo Calvino, Invisible Cities*[ii]

*"You take delight not in a city's seven or seventy wonders, but in the answer it gives to a question of yours."*
*- Italo Calvino, Invisible Cities*[ii]

When you begin your first step, you will feel a zephyr caresses your face; the same breeze sways the bamboo leaves surrounding the colourful monasteries. In the distance, The Himalayas are veiled in mist. Scents of sun-kissed rhododendrons and orchids along with the rain that drizzles down following the monsoon simultaneously stimulate your senses. You are greeted by rosy-cheeked faces as you watch them ascend the paths, maybe to Kalidaras, temple of Kali or the Tharpa Choling Monastery that has been blessed by the Dalai Lama. This is daily life in Kalimpong, a town in the Himalayan foothills by the border of India, Nepal, and Tibet. When you learn its history, you will be surprised at how a small, remote area actually holds long and rich traces of people and stories. This is an experience that one gets from wandering, exploring, and discovering. This is an addictive sensation craved by explorers of the past and present. This is ecstasy.

What if an artist contains all this sensation into a hallucinatory journey through virtual reality? We increasingly encounter virtual reality in contemporary art presentations. As we know, our brains create reality based on visual stimuli and external sensory information, thus reality can be easily deceived by optical illusion. Virtual reality can optimize visual optics to produce a sense of presence, transporting the user to a different setting and experience. Such sensation and experience of being present stretch the potential of this technology as a medium to educate and preserve cultural heritage. After all, wouldn't it be more exciting to directly experience the daily lives of ancient Romans instead of merely seeing the ruins of the civilization? In the context of art, this sense of presence offered by virtual reality also draws more artists to explore this medium.

"*Gesamtkunstwerk*", the aesthetic ideals yearned by composer Richard Wagner, where an artwork shall work to trigger and satisfy all senses, might just be achieved through virtual reality.

Artist Shezad Dawood incorporates this technology and transfers his experience in Kalimpong into a virtual reality installation, also titled *Kalimpong*. In this work, he rebuilds iconic buildings and landscapes of *Kalimpong*. Dawood seems aware of the cultural preservation narrative that can be enabled through the medium, however *Kalimpong* is not made within that paradigm of digital conservation. Despite the very detailed illustrations, *Kalimpon*g as a virtual reality artwork is not a replica of the town. Instead, this is a virtual city built from local symbols and folklores, and more importantly, the artistic perception of Dawood himself. Therefore *Kalimpong* must be understood as a universe of its own that blends together facts and fiction, integrating research and imagination. In Dawood's imagination, *Kalimpong* is the Shambala, the perfectly harmonious mythical kingdom in Tibetan Buddhism tradition. I do not intend to disclose too many artistic details of *Kalimpong*, but I may summarise the experience as both serene and isolating. In their virtual reality version, both the Morgan House, a mansion built during the British Raj era – now operates as a hotel, and the Himalayans give the impression of location settings for psychological films like The Shining or *Black Narcissus*. Yet, while we can only be a spectator of those films, in Kalimpong we can be the main actors who are present and have significant roles in exploring the city. With each scene, Kalimpong evokes some spiritual senses, to an extent, it appears as a "*mysterium tremendum et fascinans*" – both fascinating and frightening, simultaneously alluring and eerie.

The audience can see Shezad Dawood's *Kalimpong* as time capsule, where personal memories are bonded to historical facts and human experiences are metaphorically reimagined into a city, like the traveller's tales in Italo Calvino's *Invisible Cities* or the moral conflicts in St. Augustine of Hippo's *The City of God*. In both books, cities symbolize the journey of the human's spirit, where the soul wanders or eventually settles. When I talked to Shezad Dawood, I discovered his interests in Tibetan Buddhism. The cosmology and theology of Tibetan Buddhism has an emphasis on life and death summarised in the concept of "bardo". Bardo, to put it simply, is a transitional or liminal state between death and rebirth. Dreams that we have when we are asleep can also be taken as bardo. In a dream-state, we often encounter things we are familiar with and yet, everything is beyond our control; similarly, when we die nothing is in our conscious control. Although all living things shall perish, Tibetan Buddhism perceives death as an experience that is unique to each individual. A soul's journey on bardo will depend on how the person controls their memories/dreams and how they prepare themselves in the past, when they were alive. Whether death becomes painful or liberating, is conditional to each person's own ability.



**Further reading:**

i Davis, Erik. 2015. TechGnosis: Myth, Magic, and Mysticism in the Age of Information. Berkeley: North Atlantic Books.

ii Calvino, Italo. 1997. Invisible Cities. Translated by W. Weaver. London: Vintage Publishing.

I am more inclined to imagine Dawood's work as a machine of meditations, or in his own words "dream machine", a kind of bardo-state that is reinterpreted through technology to train our consciousness. Technology is very rarely placed together with mysticism and spiritualism. However, throughout human history, especially in the early civilizations, humans have always been in need of some "aid" to reach the highest consciousness or enlightenment. As an example, the priests and priestesses of Dionysus and Apollo in Ancient Greece needed opium-based hallucinatory substance to dialogue with the gods. Erik Davis, in his cult-classic *TechGnosis* combines the terms technology and gnosticism (that signifies mysticism), to demonstrate how humans have adopted the latest technologies, from cyberspace to virtual reality, to search for spiritual meanings. Simulation games and fictions presented on computers, cyberspace, and virtual reality have succeeded in annihilating the barrier of disbelief and creating a magical universe or experience, akin to the roles of shamans and priests in the ancient times. According to Davis, if this potential is fully realized, perhaps we will find new ways to pray and meditate, as long as these technologies correspond to the unabating human desire for narratives, rituals, social relations, and symbols.

I interpret the experience with *Kalimpong* as a kind of personal experience and reflections on consciousness described by ancient Chinese philosopher Zhuangzi. One day, Zhuangzi fell asleep in a meadow. As he slept, his companions saw a butterfly fluttered around the philosopher. Not long after, Zhuangzi was awakened and the butterfly disappeared. Zhuangzi smiled, he recalled, "I had a strange dream. In my dream, I turned into a butterfly, fluttering and perching on flowers." His companions told him about the butterfly that hovered around before. Zhuangzi's face brightened, he came up with his famous parable, "Ah he didn't know if he was Zhuangzi, who had dreamt he was a butterfly, or a butterfly dreaming that he was Zhuangzi."

To be conscious about perceptions is indeed something metaphysical.

# CHAPTER V
# WEARABLE TECHNOLOGY

## *Curatorial Essay*

### Wearable Technology and Speculative Fashion
**Ratna Djuwita**

Compared to other design objects and cultural products, the things that we wear are perhaps the most universal as well as the most intimate. As a cultural product, fashion is inseparable from its historical journey. Clothings of the Western culture have expanded and been adopted globally since the dawn of modernism and technological inventions in the 18th century. Since then technology helps produce a new media to see another corner of the world through fashion and sartorial choices. With Synflux's '*Algorithmic Couture*', my aim is to present a kind of "sartorial speculation" on an alternative form of fashion, which blends fashion design and wearable technology. Such speculation attempts to address, experiment, and challenge the current fashion industry by intervening in the design process, making it more in line with the relevant technological, environmental and social issues. In the long run we shall be able to anticipate the future of wearable technology and sustainable fashion. Fashion, as something that we wear is not just something that can only be associated with identity, body images or sexuality, rather fashion is an art form, a social statement, and a discourse that develops along with the progress of humankind.

# Synflux

ARTIST STATEMENT

Algorithmic Couture aims to democratize haute couture customization culture prevalent in the 19th century by revitalizing how we fashion our own style through personalization in the digital design process. As the demand for clothing increases to correspond with the growing human population, the waste created in the fashion industry continues to pile up. The existing linear model created on the premise of mass production and consumption desperately calls for a change. Looking to a more sustainable future, we must reconsider the holistic cycle of fashion. There is a need to realign our incentives to more sustainable values, by looking at how we design in fashion. By utilizing 3D scanning technology alongside CAD software, Algorithmic Couture optimize garments to the unique body types of the user, independent from the prêt-à-porter (ready-to-wear) system. This service utilizes machine learning algorithms to generate an optimized modular patterned garment allowing for the user to customize the form, material, and color to match their personal style. The optimized pattern cutting techniques that are generated are comprised of rectangles and straight lines, allowing for optimization of the fabric placement through the use of genetic algorithms. Through redesigning the creative process in how we design fashion patterns, much of the fabric wasted during the process can be reduced. By reallocating the resources that would otherwise become waste, we envision a more symbiotic relationship with machine learning in the fashion design process.

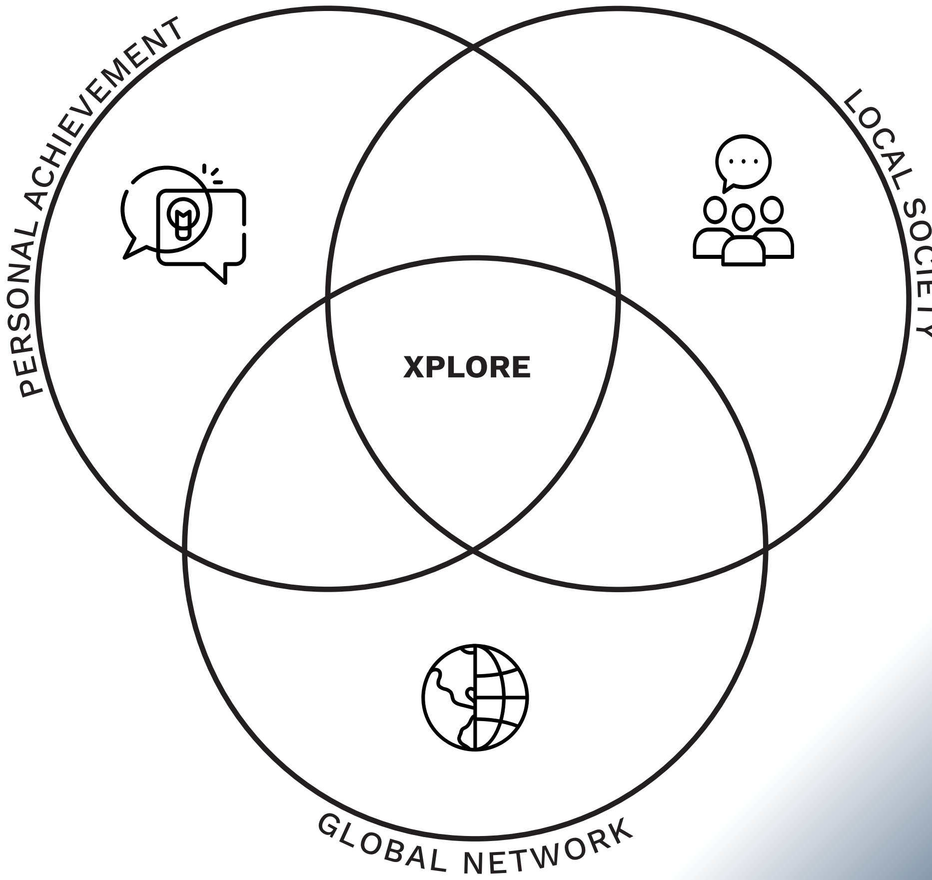


Algorithmic Couture
Cotton fabric, polyester fabric
55cm x 55cm x 180cm each (4 pieces)
2018

# XPLORE: NEW MEDIA ART INCUBATION

## *About the Program*

Initiated by ARCOLABS and HONF Foundation in 2018, *XPLORE: New Media Art Incubation* aims to support emerging artists of the new media art scene in Indonesia and Southeast Asia by equipping them with the skills and networks needed to contribute solutions to current and future challenges. The program constitutes three main activities, namely mentorship, mini residency and pitching, and all activities are designed based upon each year's theme: Art, Science and Technology (2018) and Eco-Data (2019).



**Mentorship**
A series of workshops and acquaintaces with field experts allows the participants to acquire new skills and extensive thinking abilities for their creation of new media art

**Mini Residency**
Direct associations and communications with local residents and art society fosters the participants understanding of new culture

**Pitching**
Presentation to the internationally acknowledged new media artists and curators extends opportunities and networks

## *ECO-DATA*

The law of nature has been altered by scientific and technological innovation while human body and environment have been manipulated in the favour of urban development. The consequence of exploitation of the nature for centuries has finally left us with our anxiety of dystopia and as a response, many artists have made genuine efforts to reflect the ecological threats conducted by ourselves.

This section of the exhibition presents the artistic research outcome by five young Indonesian artists who participated in *XPLORE: New Media Art Incubation* program in 2018 and 2019. Eco-Data as the chosen theme for this exhibition began with the belief that art can perform as a partner of ecological actions and agent of community changes. Acknowledging the environmental threats and its impact to all species, five emerging artists explore the way in which our life is related to the environment. In doing so, the artists pay attention to a variety of ecological problems, including pollution, animal extinction, exploitation of land; city and sanitation; and public education in regards to the environment.





# Agung Eko Sutrisno


This project is the first initiation of a long-term plan to research on the changing area of Citatah Karst region in West Java and how it affects the surrounding locals. Most of the area is filled with mining ventures that employ locals to mine and produce karst-based products like chalk. In time, with black smoke coming out of factory chimneys and endless mining, the damage of the natural landscape of Citatah is evident, some areas have in fact been wiped out completely to the ground. The artist conducts site-specific research and performance to imagine how the karst slowly disappears, how fresh air is replaced by industrial smoke, and how Citatah locals will remember this area in the future.



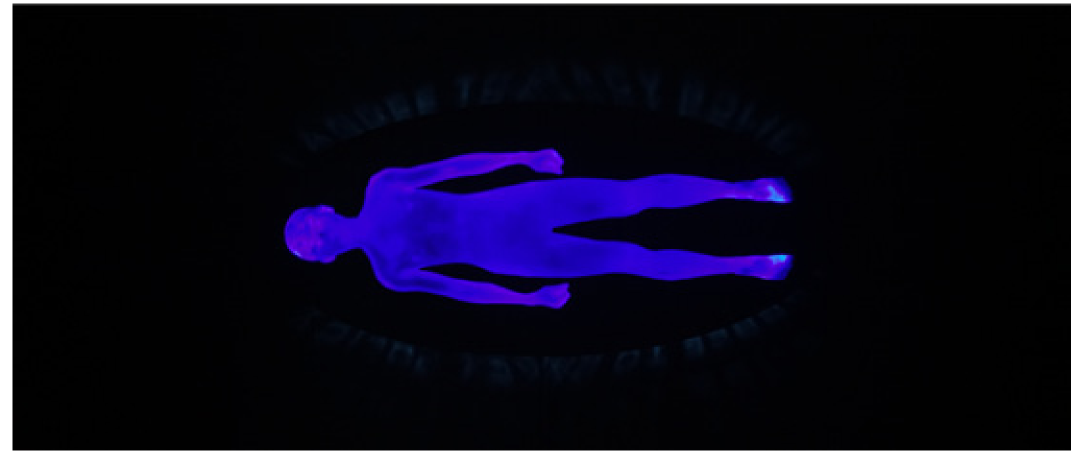
A Tiger Who Stares The Sunset
Performance video 30 minutes
Karst gypsum, iron, speaker, mp3 player
30 minutes
2019

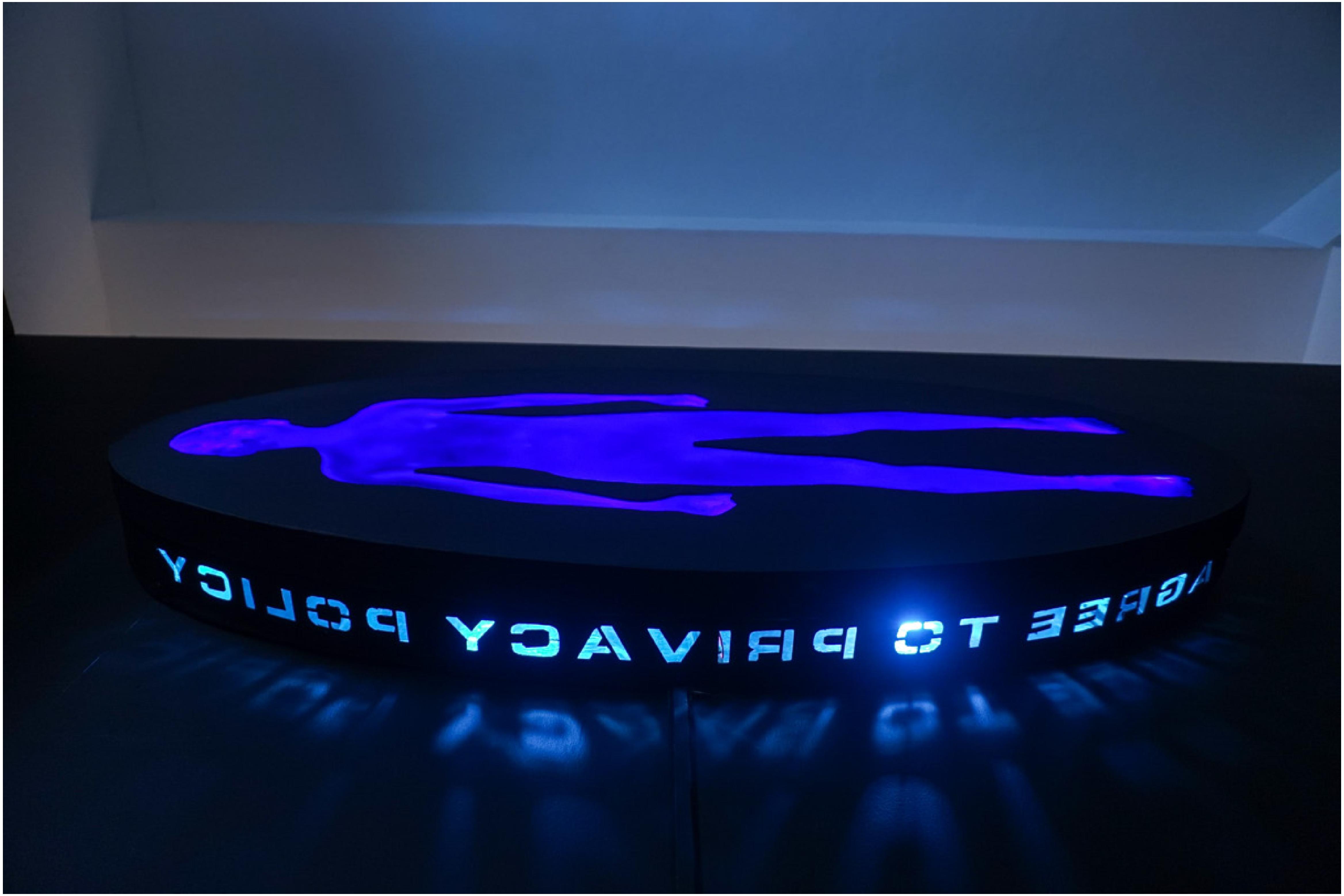
# Ariel Muhammad Zeian

ARTIST STATEMENT

Social media is an online media where users can take part, share, and create networks. Currently, it is a common media of communications used by people from various walks of life. Social media platforms demand data from the users such as name, date of birth, place of residence, and others as a requirement to create an account. Users voluntarily provide such data, perhaps unaware that all this information is stored in a database that can be analyzed almost accurately. In addition, such private information is vulnerable to be abused by certain parties. This artwork responds to the social media phenomenon in our society. The image of the body represents private identity that is transparent and accessible by the public through social media. Maybe this is a paranoid outlook on systematic advancement, but this work aims to present a different point of view on data, humans, and technology.





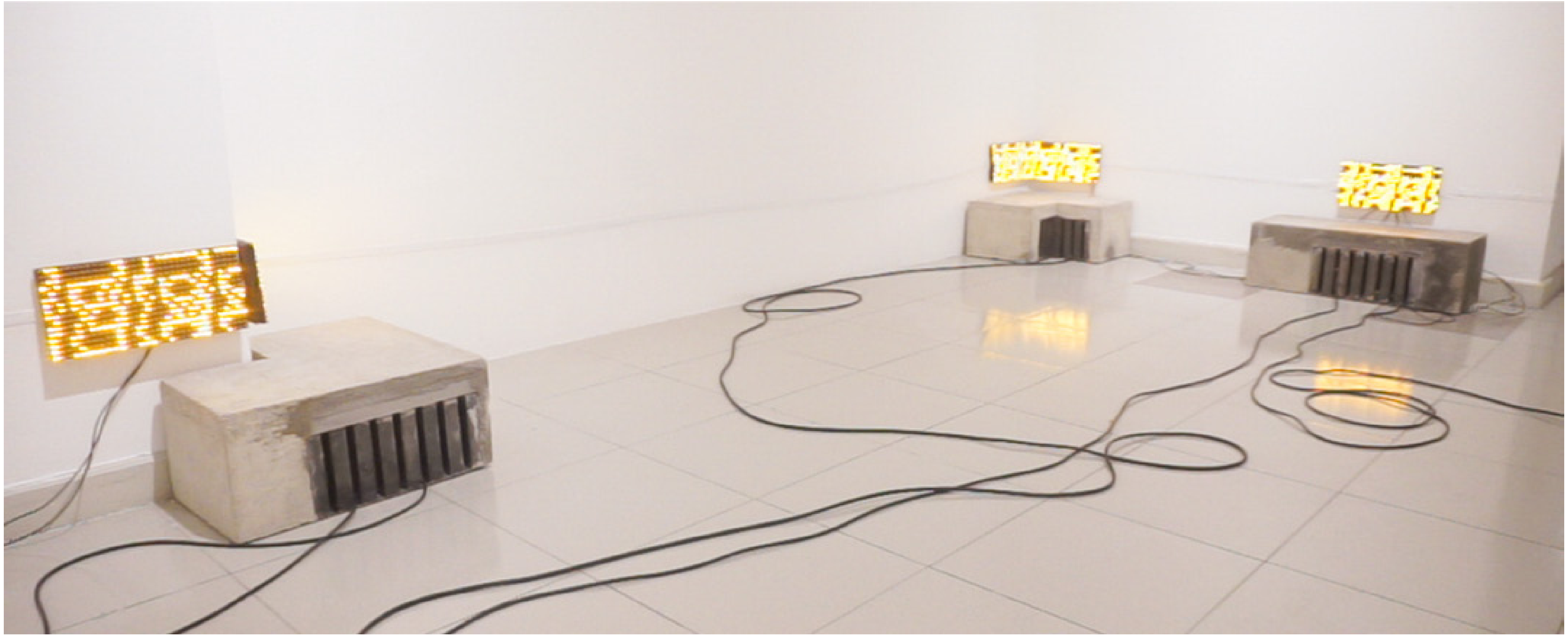

AGREE TO PRIVACY POLICY

Private (Public) Data
Neon box, LED strip, Arduino
40cm x 160cm
2019



# Delpi Suhariyanto

Water is an essential resource in every aspect of human life, including in sanitation. Yet, many Indonesians still overlook this, as evident in sanitation and sewer ducts that are clogged with plastic waste, blocking the flows of water. The artist proposes a question on the issue, just how far has technology produced a positive impact to the environment? This artwork is a sound installation that gathered specific data from the sewage system in Yogyakarta that is clogged by plastic waste. Taking the visual appearance of drainage channels by the street sides that are filled with electric cables, this artwork uses a distance sensor that encourages the audience to closely listen and reflect on the long term effects of plastic use and pollution towards sustainable living.



Upper-Ground Stream
Sewer gate, eterna cable, distance sensor,
sewer sound, LED running text
Variable dimension
2019

# M. Haviz Maha

Seventy percent of Indonesian territory is comprised of seas. However, the marine ecosystems are increasingly threatened due to pollution that happens and ends up in the seas. The artist attempts to raise awareness on the impact of pollution towards marine ecosystems and nurture empathy for marine lives, especially among children who are the next generations. This project results from the artist's teaching project, where he taught children age 5-12 in West and East Jakarta about sustainability. In the project, children were asked to create drawings of marine animals and share their imaginations about the wildlife in the polluted Indonesian seas. The artist then turned these drawings into an interactive digital simulation that visualizes the impact of pollution towards the marine ecosystem. Empathy becomes a key element as the foundation for caring and direct actions. This simulation aims to be an early education tool about sustainability to initiate behaviour change towards a more environmentally-friendly lifestyles.

## ARTIST STATEMENT

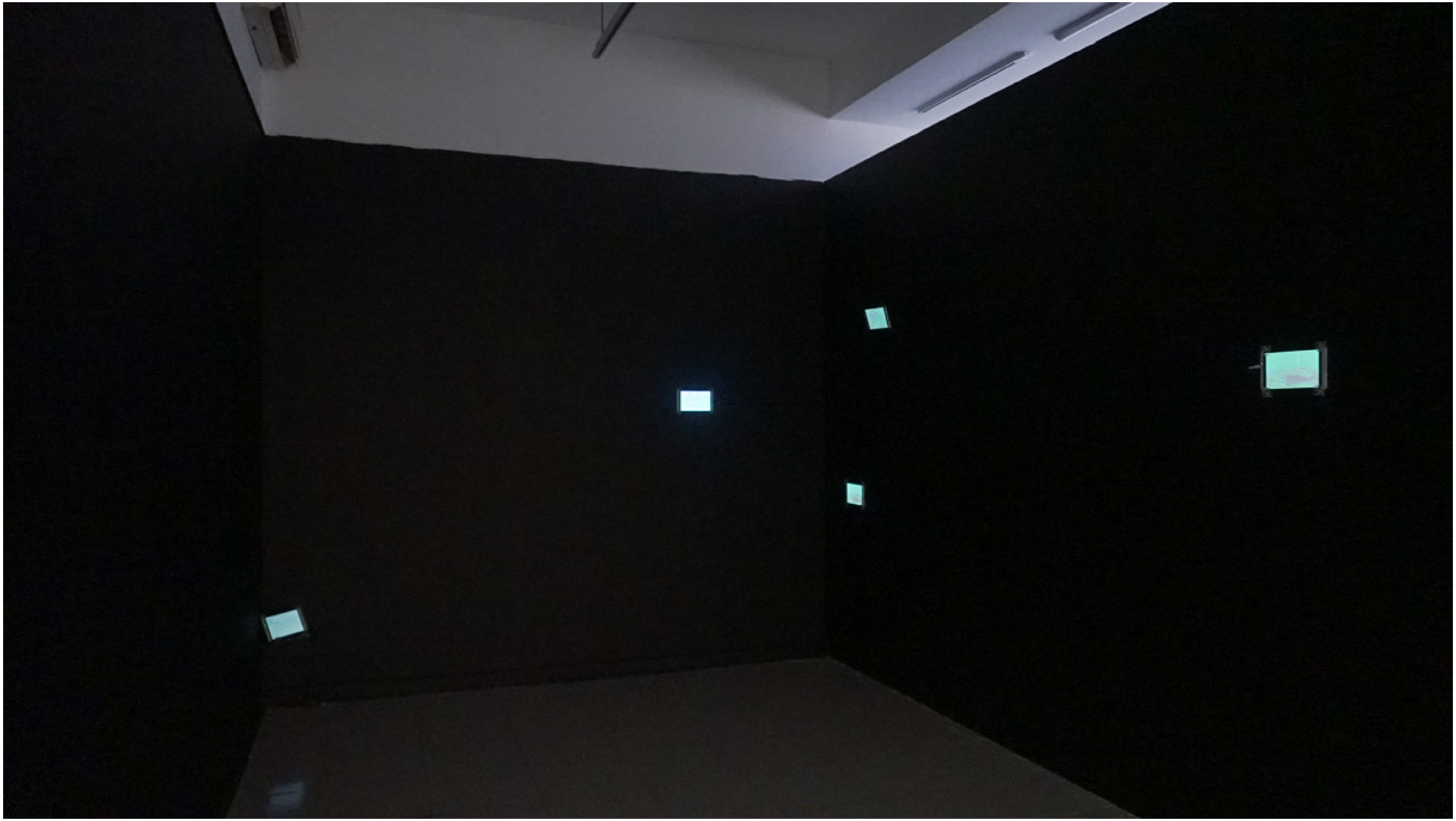
#EducationProject SeaofEmpathy
Video documentation, water pool, projection
Variable duration
2019

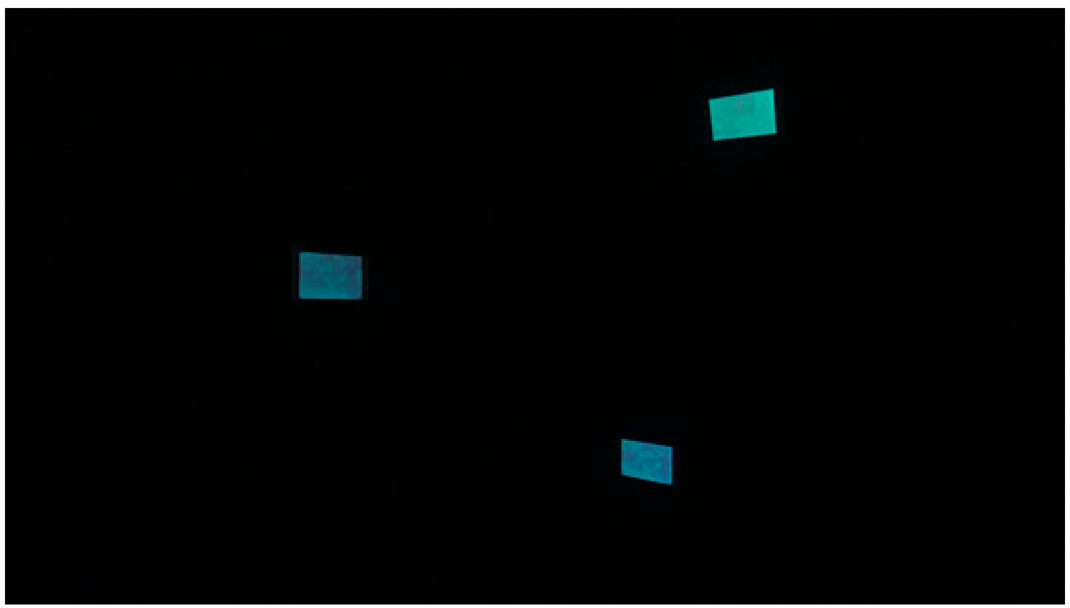


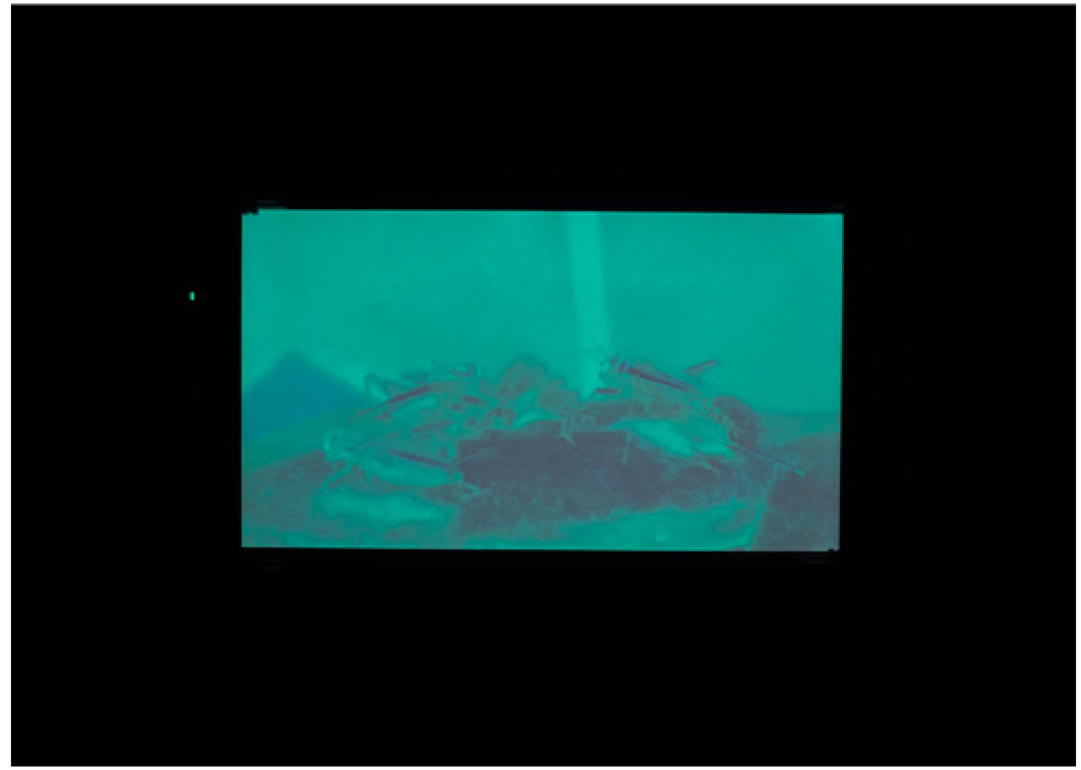


# Mira Rizki

As time goes by, ecosystems in this world develops according to mutual relationship between humans and the environment. As fellow inhabitants of nature, humans, animals, and plants produce an ongoing dialogue that affects life on the planet. The current state of our environment is a reflection of its inhabitants, predominantly humans, who create the biggest impact and dominate the ecosystem. The sounds of the industrious human civilization have intervened and disrupted the tones and sounds of nature. Animalresident is a sanctuary of artificial sounds of man made creations, collected from the urban environment. The resemblance of this urban sounds with wild animals is a longing for mother nature and the sounds of its children.





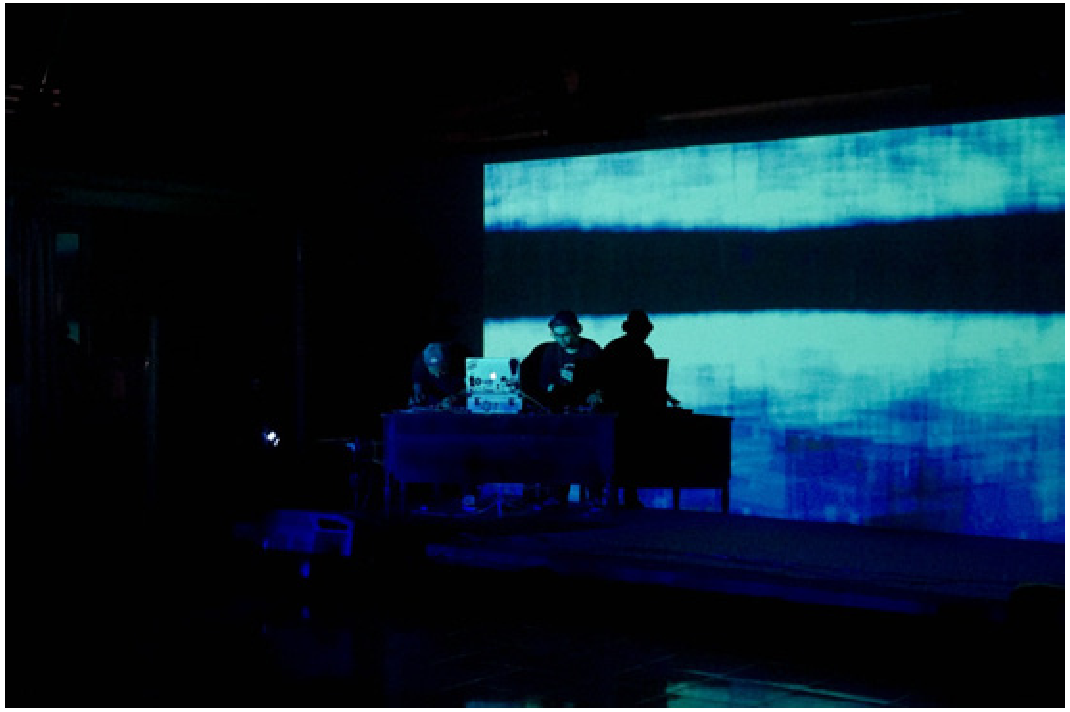


Animalresident
6 channel video installation
Variable duration
2019

## *Agenda*

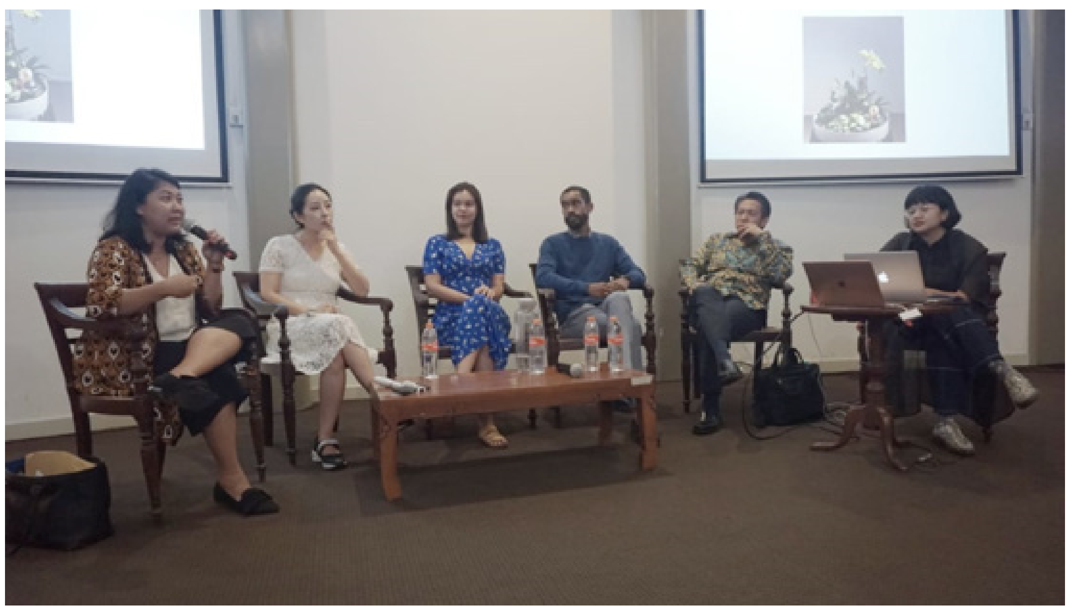
**Exhibition Period**
October 21 – November 7, 2019

**Monday, October 21**
Press Tour: 16:30pm-18:00pm
Opening Ceremony: 19:00pm-22:00pm
Officiated by Tubagus ‘Andre’ Sukmana
Music Performance by Pasukan Musang




**Tuesday, October 22**
Seminar: 13:00pm-15:00pm
"Art, Science and NAture: Towards Future Possibilities"
Speakers: Aprina Murwanti, Cho Eun Woo, Dharsono Hartono, Digital Nativ
Moderator: Nin Djani



**Saturday, October 26 & November 2**
Gallery Tour: 13:00pm-14:00pm




## Cho Eun Woo
Artist

Cho Eun Woo (b. 1981) is currently a Chief Branding and Design Officer at Robo Risen. Cho graduated from OCAD University in Toronto and completed a graduate program at SVA New York City. Later, she certified her MPhil/ PhD. in Art and Philosophy at IDSVA. Upon arrival in Korea, she presented artworks related to science and brain waves, attempting to introduce this new theme of art and expand her artistic repertoire. Her notable art exhibitions and projects include Reina Sofia National Museum Spain - Madrid, Les Rencontres Internationales Paris, Beaux-Arts de Paris, and the Korean Cultural Center New York and Los Angeles. More recently, she displayed her works at the 21st Korean Science Fair, Seoul Museum of Art, Platform L, and got involved in Seoul Foundation for Arts and Culture's Seoul Power Station Project and National Gallery of Indonesia. Her artworks were invited to Global Meeting at Hyundai Motor Group, Seoul National University of Science and Technology and NYU Florence Political & Science Program.

## Digital Nativ
Artist

Digital Nativ (est. 2015) is a design and fabrication studio exploring technological processes. As a collective, the studio focuses its work to research, modify and customise industrial grade and emerging technologies. Working across technical disciplines, Digital Nativ builds configurable and modular systems that are founded on design principles to hack the gap between the industry and the individual. Notable digital systems project include Adidas, BMW, GO-JEK, Hermès, Lanvin, Longchamp and Samsung. In addition, Digital Nativ has collaborated with Invisible Flock for Nada Bumi and Aurora Project commissioned by British Council Indonesia as part of UK/ID Festival 2017.





## Eva Bubla
Artist

Eva Bubla (b. 1985) is a Hungarian artist based in Asia since 2011. She graduated in painting, yet in recent years her works have become strongly connected to the environment in an effort to nurture environmental awareness. For Eva, nature is a teacher and collaborator thus she mostly uses natural, re-used, and recycled materials in her art pieces. Her works mainly involves worldwide research on social and ecological conditions, particularly on water, farming practices, and the impact of urbanization, chemical farming, and commercial activities on natural landscape, which then serve as a basis for her installations and performances. She also creates site-specific works and often involve locals during the creation process to work together to find sustainable solutions for ecological and social problems. She is the co-founder of Green Root Lab, an organization dealing with environmental art, education, and ecological activism.

## Shezad Dawood
Artist

Shezad Dawood (b. 1974) is a London-based artist. He was trained was trained at Central St Martin's and the Royal College of Art before undertaking a PhD at Leeds Metropolitan University. He is a Research Fellow in Experimental Media at the University of Westminster. He works across disciplines: film, painting, neon, sculpture and more recently virtual reality to deconstruct systems of image, language, site and narrative. Using the editing process as a method to explore both meanings and forms, his practice often involves collaboration and knowledge exchange, mapping across geographic borders and communities. Through a fascination with the esoteric, otherness and science-fiction, Dawood interweaves histories, realities and symbolism to create richly layered artworks. His works have been exhibited internationally and is in major and public collections, including Tate, London; LACMA, Los Angeles; National Gallery of Canada, Ottawa; The British Museum, London; Kiran Nadar Museum of Art, Delhi; Mathaf, Arab Museum of Modern Art, Doha; UBS Collection; US Government Art Collection; Government Art Collection, UK; Bill & Melinda Gates Foundation; Art Jameel, Dubai; Devi Art Foundation, Delhi; EKARD Collection.

## Synflux
Artist

Synflux (est. 2018) is a Tokyo-based research collective seeking new paradigms in fashion by focusing on design research and fashion design. Founded by engineers Kye Shimizu and Yusuke Fujihira, designer Kotaro Sano and fashion designer Kazuya Kawasaki, Synflux is a collaborative project between fashion designers, engineers, architects, researchers, and scientists. The collective aims for a more holistic and sustainable cycle of fashion by using technology to reduce fabric waste. By looking into historical, existing, and speculative references, Synflux explore the intersections between humanity, environment, and technology and how this relationship can develop in the future. Their project, Algorithmic Couture received the WIRED Creative Hack Award 2018.

## Agung Eko Sutrisno
Artist

Agung Eko Sutrisno (b. 1994) is a performance artist based in Bandung. In his works, Eko is interested in archives, historical values and preservation of collective memories. He practices site-specific performance methods and uses media technology such as video, sound, and installation as extensions of his body. Eko is the founder of prfrmnc.rar, a laboratory focusing on performance art discourses in Bandung and an active member in Bandung Performing Arts Forum (BPAF) and Klub Remaja Visual Art Artist Collective. He has taken part in various workshops, including with Seiji Shimoda in ASBESTOS Art Space, Bandung (2015) and with Melati Suryodarmo at Jakarta Biennale (2017). Among the art projects and residencies that he has participated in are, "Welcome to Jakarta", Micro Galleries Jakarta (2017), "Landscape Painting", Orbital Dago (2018), "Simpati Paradiso", Jatiwangi Art Factory (2019). Eko is one of the semi finalists for Bandung Contemporary Art Award (BaCAA) 2019





## Ariel Muhammad Zeian
Artist

Ariel Muhammad Zeian (b. 1997), also known as Ian, is a new media artist based in Malang and Pasuruan. With an art education background, he has been active in several art collectives and art initiatives in East Java. His interest in new media art started when he joined TITEN – a new media art residency program initiated by a Surabaya based collective WaftLab.  Ian's recent work is inspired by the local tradition of "oprek", which is the art of assembling different parts into one functional machine. Recently he also participated in the Media Art Week, Pekan Seni Media 2019: Sinkronik in Samarinda, East Kalimantan, organized by the Ministry of Education and Culture in Indonesia.

## Delpi Suhariyanto
Artist

Delpi Suhariyanto (b. 1996) is a Bandung-based artist who explores sound and performance. He is active on discussions about the mass media as a vulnerable tool of information distribution. Prior to this project, Delpi took part in XPLORE: New Media Art Incubation initiated by Arcolabs and HONF, where he incorporated a new media art approach to his sound installation. New media as an art medium has an intense connection with people's everyday life. Delpi sees similarities between people's attempt to create a product to make their life easier and new media art because both essentially put technology as an accessible tool to make improvements in life and to resist something. In 2019, Delpi also took part in several art projects and exhibitions, including Soemardja Sound Art Project, Soemardja Gallery, Bandung and Pekan Seni Media 2019: Sinkronik, Samarinda.

## M. Haviz Maha
Artist

M. Haviz Maha (b. 1993) lives and works in Jakarta and Tangerang. He graduated from the Visual Art Education at the Jakarta State University (UNJ). His interest in media and technology-based art began when he joined as a videographer and animator at Serrum, an artist collective focusing on art education. In 2016, he and his friends founded Sinema Kolekan, a research unit in the student art festival Jakarta 32. Through this unit, they examine audiovisual works, organize discussions and compile archives.

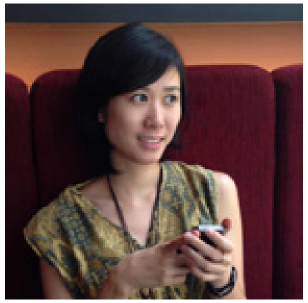
## Mira Rizki
Artist

Mira Rizki (b. 1994) is a new media artist focusing on sound and interacti-vity as her artistic practice. Her career has begun since her time as a student at Intermedia Studio, Faculty of Arts and Design, Bandung Institute of Technology (ITB). Her works can be characterized in its playful and participatory nature. With these experiments and interactions, she aims to engage the audience in the sound-making process and raise their awareness towards their surroundings. She has participated in several residency programs in Germany and Japan. In 2019, she was selected as one of the finalists on Trimatra Competition, organized by Salihara Gallery, Jakarta.





## Evelyn Huang
Curator



Evelyn Huang (b. 1984) is an Indonesian curator and lecturer who pursues knowledge as a never ending path for her life. Evelyn graduated with an MFA in Creative Entrepreneurship from Jakarta Institute of Arts after she obtained her Bachelor's Degree in Visual Communication Design, University of Pelita Harapan and Master's Degree in Cultural Studies, University of Indonesia. In recent years, she has curated EXI(S)T, one of the most renowned art incubation programs in Indonesia. Aside from her full time position as Program Head in IDS - International Design School, she also curates media art exhibitions and international exchange exhibitions for ARCOLABS.

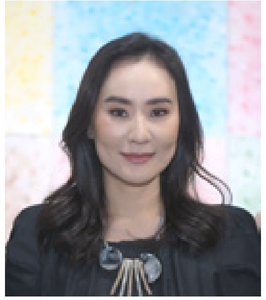
## Irine Agrivina
Curator

Irene Agrivine (b. 1976) graduated from the Faculty of Graphic Design at the Indonesian Institute of the Arts (ISI), Yogyakarta, and continued her study at the Graduate School of Religion & Cultural Studies in University of Sanata Dharma, Yogyakarta. She is one of the founding members and directors of HONF, a new media and technology laboratory based in Yogyakarta. In 2013 she co-founded XXLab within HONF as a collective of women-artists focusing on arts, science and free technology. One of their projects, 'SOYA C(O)U(L)TURE' (2015) was among the selected projects to win an award from the Prix Ars Electronica 2015 in Linz, Austria.

BIOGRAPHY



## Jeong Ok Jeon
Curator



Jeong Ok Jeon (b. 1975) is a Jakarta-based Korean curator and educator who is actively engaging in Southeast Asian contemporary art, especially working on providing international exposures for regional artists in and outside of Indonesia. Jeon worked as a curator in an alternative art space called SSamzie Space in Korea in the early 2000s; after moving to the United States, Washington D.C. Metropolitan Area in 2005, she worked as an independent curator focusing on promoting Asian art and artists in the US. Since relocating to Indonesia in 2011, she has been passionately involved in both contemporary art scene and art academy for the past nine years. With interests in new media and interactive art, she has curated numerous science and technology-based art exhibitions. The most recent project is Media Art Week (Pekan Seni Media) 2019: SINKRONIK, an annual program hosted by Indonesia's Ministry of Education and Culture. She earned an MFA from the Savannah College of Art and Design in the US and a BFA from Ewha Womans University in Korea. Jeon currently serves as the director at ARCOLABS – Center for Art and Community Management and a full time lecturer at the Department of Visual Art Education, Faculty of Language and Arts at Jakarta State University.

BIOGRAPHY

## Nin Djani
Curator


Nin Djani (b. 1992) is a Jakarta-based Indonesian curator, writer, and editor. She entered the art scene in 2015 when she worked as a communications manager at Suar Artspace, an alternative art space dedicated to Indonesian young, emerging artists. Since then she has worked as an independent curator in numerous art projects relating to interdisciplinary subjects, cultural exchange, science and technology. She holds an MA in Southeast Asian Studies from Universiteit Leiden and a BA in Media and Communications from Goldsmiths, University of London. Nin joins ARCOLABS in 2017 as a curator while maintaining her position at HOWDIY, a production company focusing on digital content, as a Creative Development Director where she helps develop narrative and visual storytelling for digital commercials.





## Ratna Djuwita
Curator

Ratna Djuwita (b. 1983) has been actively involved in House of Natural Fiber (HONF) since 2011. She initiated a project with HONF called Open Apparel, a platform aiming to create new opportunities for a secondary industry and increase employment opportunities in the Indonesian fashion industry. In 2015, with 4 women artists in Yogyakarta, she co-founded XXLab, a women collective focusing on art, science, and free technology. Their recent and ongoing project is 'SOYA C(O)U(L)TURE', a research project on soy waste. In this ambitious research, XXLab uses bacteria and tissue culture to create alternative fuels, food, and skin-like sheets from soy waste. The objective is to create sustainable production using simple technology and reducing waste that pollutes water. Currently she is pursuing her Master's at the Graduate School of Religion & Cultural Studies in University of Sanata Dharma. Aside from that she continues to develop products from her experiments with new materials she did with XXLab.

## Aprina Murwanti
Writer

Aprina Murwanti (b. 1982) has 15 years of professional experience in art education, research and management of social programs. She always tries to bring visual arts closer to the people through various strategies oriented to the distribution of knowledge, community needs and public participation. Aprina holds a PhD in Contemporary Art Practice from Wollongong University, Australia and an alumna of Bandung Institute of Technology (ITB). She currently serves as Head of Education and Public Programs at the Museum MACAN (Museum of Modern and Contemporary Art in Nusantara), a position she took in 2018, and since 2004 has worked as a lecturer at the Department of Visual Art Education at Jakarta State University.

## Rumi Siddharta
Writer

Rumi Siddharta (b. 1988) obtained a Master's in Fine Arts from Bandung Institute of Technology (ITB). Currently, he focuses his career in styling and art direction. He is always fascinated by the intertextuality of things. An avid reader in art history and a self-proclaimed history geek, Rumi often spends his free time sharing stories on film and literature via instagram, @rumisiddharta.



## Yuka Dian Narendra
Writer

Yuka D. Narendra (b.1972) is a metalhead, a big fan of Rush, Voivod and Slayer. Since graduating from the Department of Visual Communication Design, University of Pelita Harapan in 2000, he went straight to academia and became a lecturer at his alma mater. His interests in popular culture drove him to start reading Cultural Studies textbooks during university before continuing his Master's in Cultural Studies at the Faculty of Humanities, University of Indonesia from 2004 - 2006. He returned to teaching at his alma mater, as well as several other private universities. Aside from teaching, he writes and presents his research in various scientific forums. He authored and published Heavy Metal Parents, a book based on his research about the 1980s generation of metalheads in Indonesia. Yuka spends his spare time in his modest recording studio as an underground music producer, who has produced more than 20 albums with a wide range of genres from metal and grunge all the way to avant-garde jazz, pop, and folk music. Currently he is a full time lecturer of Visual Communication Design, Matana University, Tangerang. When he is not working on campus and in the studio, Yuka enjoys hunting authentic Chinese food with his wife and daughter.

## *Credits*

**Five Passages to the Future**
21 October - 7 November 2019

Galeri Nasional Indonesia
Jl. Medan Merdeka Timur. No.14
Jakarta Pusat, Indonesia

### MANAGEMENT

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National Gallery of Indonesia

**Organizer**
ARCOLABS & HONF FOUNDATION

**Treasurer**
Jeong Ok Jeon

**Sponsorship**
Evelyn Huang
Jeong Ok Jeon
Nin Djani



### EXHIBITION

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Jeong Ok Jeon
Nin Djani
Ratna Djuwita

**Gallery Liason**
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**Production Manager**
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**Media Partner & Public Relation**
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Adi Priyatna



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